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SYSTEM AND METHOD OF FABRICATING A REINFORCED STRUCTURAL MEMBER

Abstract

A structural member optimized to resist lateral forces, as well as systems and methods of fabricating and implementing same, are disclosed. In some implementations, a modular framework erected at a work site may support forms used for the fabrication of reinforced concrete or masonry structural members which do not have uniform cross-sectional thickness. The modular nature of the framework allows a construction crew to arrange the forms quickly and precisely on site, and the varying cross-sectional thickness of the structural member and positioning of reinforcing elements economize on materials usage while simultaneously increasing the structural member's resistance to lateral loading.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable.

FIELD OF THE DISCLOSURE

[0002] Aspects of the disclosed subject matter relate generally to structural members of a building, and more particularly to a reinforced structural member that is optimized to resist lateral forces in addition to vertical forces, as well as to systems and methods of fabricating and implementing same.

BACKGROUND

[0003] Concrete structures have been known and widely used for millennia, dating back at least to the early days of the Roman empire. Some ancient structures built by the Romans stand today, due partially, though in no small part, to concrete's ability to withstand compression very well. While tolerating compressive loads well enough to support great weight in an axial direction, concrete is less robust under shear, flexure, and torsion. In modern construction design, time-tested concrete materials and recipes are routinely enhanced by inclusion of structural reinforcing elements (such as metal “REinforcing BARs,” or “rebar,” metal meshes, or other fibrous materials) that may be embedded into the concrete structure as the concrete is allowed to cure; in practice, these metal bars, meshes, or other reinforcing elements are employed to support the concrete under shear, tensile, or torsional loads. In some instances, the reinforcing elements themselves may be subjected to tensile loads as the concrete is curing, and the tension is removed when the concrete has set—this is typically referred to as “pre-stressing,” and places the concrete under compression when the structural element is completed, making the overall structure generally more resistant to tension during use.

[0004] Typical standards for reinforced concrete structures, however, encourage wasteful construction practices assuming, for safety and other purposes, that it is necessary to support a concrete wall, for instance, that is subjected to tension in two opposed directions. Specifically, engineering best practices suggest that a designer consider lateral loads on a wall from two principal horizontal directions: normal to an exterior surface of the wall from the outside in; and normal to an interior surface of the wall from inside out. With these competing interests in mind, building codes and construction professionals have heretofore compromised by placing applicable reinforcing elements such as rebar or other materials at the midpoint of such a wall (i.e., midway between the exterior surface and the interior surface).

[0005] Traditional construction paradigms use more concrete than is necessary to support the vertical loads to be expected for a given application, and merely position the reinforcement elements where they provide sufficient resistance, but not optimal resistance, to lateral loads expected for that application. A more elegant solution would be to economize on building materials while engineering the wall or other structural member for optimal resistance to expected lateral loads given the reduced materials requirements.

[0006] Though there is a continuing need for improved systems and methods of fabricating structural members for building design and construction applications, there is no current commercially available solution that maximizes resistance to lateral loads while simultaneously minimizing materials requirements and work crew efforts that are necessary to satisfy a given set of performance parameters. Such a solution may not only provide structural members that satisfy specific design and load resistance criteria, but also may reduce resources and time required by construction crews in the field to construct such structural members.

SUMMARY OF THE DISCLOSURE

[0007] The following presents a simplified summary of the disclosure in order to provide a basic understanding of some aspects of various embodiments disclosed herein. This summary is not an extensive overview of the disclosure. It is intended neither to identify key or critical elements of the disclosed embodiments nor to delineate the scope of those embodiments. Its sole purpose is to

present some concepts of the disclosed subject matter in a simplified form as a prelude to the more detailed description that is presented later.

[0008] The present disclosure describes a system and method employing a modular framework to support forms used for the fabrication of reinforced concrete or masonry structures. In the context of this description, such a system and method leverage the modular nature of the framework to arrange the forms quickly and precisely. In some implementations, for instance, the forms may be selectively suspended from a rail assembly that may be adjusted to be level and plumb independent of the surface upon which the support framework is attached, such that inevitable slight undulations or other imperfections in a foundation, floor joist system, or other supporting element may not influence the precise placement and relative orientation of the formed structure. It will also be appreciated that the construction of the structural member itself is unique, independent of the method system employed to fabricate it.

[0009] In accordance with one aspect of the disclosed subject matter, for example, a structural member may generally comprise: a first surface oriented to be substantially vertical, having a first height, and positioned to be exposed to an exterior side of a building; a second surface oriented to be substantially vertical, having a second height equal to the first height, and positioned substantially parallel to and in a spaced-apart relationship with the first surface, the spaced-apart relationship defining a substantially uniform baseline thickness of the structural member; a recessed region formed in a portion of the second surface and extending toward the first surface, the recessed region having a recess height that is less than the second height, a recess width that spans a horizontal distance of the second surface, and a recess depth from the second surface that is less than the baseline thickness of the structural member; wherein the recessed region defines an effective thickness of the structural member that is equal to a difference between the baseline thickness and the recess depth; and a reinforcing element intermediate the first surface and the second surface; wherein the reinforcing element is disposed intermediate the recessed region and the first surface, supports a portion of the structural member having the effective thickness, and is selectively positioned relative to the first surface such that a resistance to a given bending moment for the structural member as a whole is optimized.

[0010] Structural members are disclosed wherein a material used for the structural member and defining the first surface and the second surface is concrete. In some implementations, the reinforcing element may be embodied in or comprise a steel bar, a lattice of interconnected steel bars, a web structure, or a combination of these. Implementations are disclosed wherein the web structure comprises a steel mesh, a fibrous material, or a combination of these or other materials.

[0011] Systems are disclosed wherein the recessed region is symmetrical in planar cross-section. In some circumstances, the recessed region is rectangular in planar cross-section, though other alternatives are also disclosed. For example, the recessed region may be trapezoidal or curved in planar cross-section.

[0012] In accordance with another aspect of the disclosed subject matter, a method employing a modular framework system to facilitate fabrication of a structural member may generally comprise: securing an elongate base plate to a foundation operative to support the structural member, the securing comprising selectively attaching the elongate base plate to the foundation in an orientation substantially perpendicular to a longitudinal axis of a footprint of the structural member; attaching a column to a portion of the elongate base plate proximate to the footprint of the structural member and bracing the column in a substantially vertical orientation using a brace member attached to a portion of the elongate base plate distal to the footprint of the structural member, the column having a height selected to accommodate a selected height of the structural member; attaching a top support to the column opposite the elongate base plate and securing the top support in an orientation substantially parallel to a direction of the elongate base plate; selectively attaching a rail assembly to the top support in a substantially horizontal orientation and substantially parallel to the longitudinal axis of the footprint of the structural member; selectively suspending an interior form

element from the rail assembly to define an interior surface of the structural member; selectively suspending an exterior form element from the rail assembly a distance from the interior form element to define an exterior surface of the structural member, the distance between the interior form element and the exterior form element defining a baseline thickness of the structural member; optionally selectively suspending an opening form element to define an opening, in which no structural material will be present, in the structural member, the optionally selectively suspending comprising selectively attaching the opening form element to at least one of the interior form element or the exterior form element; selectively suspending a reinforcing element from the rail assembly intermediate the interior form element and the exterior form element and positioning the reinforcing element in a position relative to the exterior form element to support the structural member against a bending moment; and depositing a slurry of construction material in a cavity between the interior form element and the exterior form element to form the structural member having the selected height.

[0013] Methods are disclosed wherein the selectively suspending an interior form element comprises suspending a recess element, having a recess depth from the interior surface of the structural member, that creates a recessed region having an effective thickness that is less than the baseline thickness of the structural member when the slurry is deposited in the cavity.

[0014] As noted above in connection with the structural member, methods are disclosed wherein the recess element may be symmetrical, rectangular, trapezoidal, or curved in planar cross-section.

[0015] In some disclosed methods, the selectively suspending a reinforcing element comprises positioning the reinforcing element proximate to the external form element such that the reinforcing element is entirely encased within an area of the structural element having the effective thickness.

[0016] Methods are disclosed wherein the depositing a slurry comprises depositing concrete in the cavity. In that regard, some such methods may further comprise utilizing a trolley that translates on the rail assembly along the longitudinal axis of the footprint of the structural member to deposit the concrete (or other construction slurry) in the cavity.

[0017] In accordance with yet another aspect of the disclosed subject matter, a modular framework system to facilitate fabrication of a structural member may generally comprise: an elongate base plate selectively attached to a foundation, which is operative to support the structural member, in an orientation substantially perpendicular to a longitudinal axis of a footprint of the structural member; a column attached to a portion of the elongate base plate proximate to the footprint of the structural member and having a height selected to accommodate a selected height of the structural member; a brace member attached to the column at a location above the foundation and below the height of the structural member and to a portion of the elongate base plate distal to the footprint of the structural member such that the brace member braces the column in a substantially vertical orientation during use; a top support attached to the column opposite the elongate base plate in an orientation substantially parallel to a direction of the elongate base plate; a rail assembly attached to the top support in a substantially horizontal orientation and substantially parallel to the longitudinal axis of the structural member; an interior form element selectively suspended from the rail assembly and defining an interior surface of the structural member; an exterior form element selectively suspended from the rail assembly a distance from the interior form element to define an exterior surface of the structural member, the distance between the interior form element and the exterior form element creating a cavity that defines a baseline thickness of the structural member; an opening form element optionally selectively suspended from at least one of the interior form element or the exterior form element to define an opening, in which no structural material will be present, in the structural member; and a reinforcing element selectively suspended from the rail assembly intermediate the interior form element and the exterior form element and positioned relative to the exterior form element to support the structural member against a bending moment.

[0018] Systems are disclosed wherein the interior form element comprises a recess element, having

a recess depth from the interior surface of the structural member, that creates a recessed region having an effective thickness that is less than the baseline thickness of the structural member when a slurry is deposited in the cavity. In some such systems, the recessed region is symmetrical in planar cross-section. By way of example, systems are disclosed wherein the recessed region is rectangular, trapezoidal, and curved in planar cross-section.

[0019] It is noted that various aspects and features of some disclosed implementations may be employed readily with other aspects and features disclosed in connection with other implementations. The foregoing and other aspects of various disclosed embodiments will be apparent through examination of the following detailed description thereof in conjunction with the accompanying drawing figures, in which like reference numerals are used to represent like components throughout, unless otherwise noted.

Description

DESCRIPTION OF THE DRAWING FIGURES

[0020] FIG. 1A is a simplified side perspective view of a prior art reinforced concrete wall;

[0021] FIG. 1B is a simplified vertical cross-sectional view of the prior art wall of FIG. 1A, taken along a transverse axis illustrated in FIG. 1A;

[0022] FIG. 1C is a simplified side perspective view of one implementation of a reinforced structural member according to one aspect of the disclosed subject matter;

[0023] FIG. 1D is a simplified vertical cross-sectional view of the reinforced structural member of FIG. 1C, taken along a transverse axis illustrated in FIG. 1C; FIG. 1E is a more detailed side perspective view (from the opposite side) of the reinforced structural member of FIG. 1C showing a span having a baseline thickness between adjacent spans having a thickness that is less than the baseline thickness;

[0024] FIGS. 2A through 2E show various elevation, side, and perspective views of a base plate and a brace shoe for use in connection with a system for forming a reinforced structural member;

[0025] FIGS. 3A and 3B show, respectively, a side view and a plan view of a column and brace members for use in connection with a system for forming a reinforced structural member;

[0026] FIGS. 4A through 4C show a side view of a top support structure, and plan views of two implementations of a top support structure, for use in connection with a system for forming a reinforced structural member;

[0027] FIGS. 5A through 5C show a side view and plan views of a rail assembly attached to the top support of FIGS. 4A through 4C and operative for use in connection with a system for forming a reinforced structural member;

[0028] FIGS. 6A, 6B, and 6D through 6O show various side and plan views of implementations of a form selectively suspended from the rail assembly of FIGS. 5A through 5C, or from other forms, and operative to define an interior surface of a reinforced structural member;

[0029] FIG. 6C is a plan view of a lower rail assembly operative to secure a form to a lower portion of a support column as best illustrated in FIGS. 6A, 6E, 6K, and 6L;

[0030] FIGS. 7A and 7B show, respectively, a side view and a plan view of aspects of a reinforcing system for use in connection with a reinforced structural member;

[0031] FIG. 8A shows a side view of one implementation of a form selectively suspended from the rail assembly of FIGS. 5A through 5C and operative to define an exterior surface of a reinforced structural member;

[0032] FIG. 8B shows a side view of one implementation of a trolley supported by the rail assembly of FIGS. 5A through 5C and operative to deposit a slurry of construction material to form a reinforced structural member; and

[0033] FIG. 9 is a simplified functional flow diagram illustrating aspects of one implementation of

a method of fabricating a reinforced structural member.

DETAILED DESCRIPTION

[0034] Certain aspects and features of the disclosed subject matter may be further understood with reference to the following description and the appended drawing figures. In operation, a system and method of fabricating a reinforced structural member may have utility in connection with various construction projects for buildings, dwellings, and other enclosures. Specifically, the present disclosure provides for construction systems and methodologies that may generally make use of conventional hardware and tools that are familiar to construction contractors and work crews, but do so in a manner that is specifically designed to conquer some of the challenges (that have heretofore been unaddressed) associated with typical construction project practicalities.

[0035] The following detailed description and the appended drawing figures describe and illustrate some implementations of a system and method for the purpose of enabling one of ordinary skill in the relevant art to make and use these implementations. As such, the detailed description and drawing figures are purely illustrative in nature and are in no way intended to limit the scope of the disclosure in any manner. It should also be understood that the drawing figures are not necessarily to scale and that in certain instances, details which are not necessary for an understanding of the disclosure, such as details of material hardness or strength and curing times or other parameters, may have been omitted from the drawing figures or the written description for clarity. As noted above, in the accompanying drawing figures, like reference numerals are used to represent like components throughout, unless otherwise noted.

[0036] The features of some implementations are described below with reference to illustrations of methods, systems, and structural members, per se, according to the disclosed implementations. As noted above, it will be understood that individual blocks or structural components depicted in the drawing figures, as well as certain combinations of blocks depicted in the drawing figures, may be facilitated by computer program instructions, hardware (electrical, electro-mechanical, or mechanical) devices, or a combination of both, used in connection with automating aspects of the method, such as, for instance, design, manufacture, placement, and alignment of physical components of the system. Some computer program instructions may be provided to or from a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce or enable a particular machine, such that the instructions, which are executed via the processor of the computer or other programmable data processing apparatus, implement the functions specified in the block or blocks, either individually or in combination. Such computer systems may be used to design, arrange, and/or fabricate the physical components depicted in the drawing figures or their equivalents. This will be especially apparent in the discussion of the placement of physical components which embody the system or parts thereof.

[0037] In that regard, it is noted that aspects of the present system and method may be designed, tested, refined, modified, or implemented with the assistance of or in connection with one or more computing devices, including one or more servers, one or more client terminals, including computer terminals, a combination thereof, or on or facilitated by any of myriad computing devices currently known in the art, including without limitation, personal computers, laptops, notebook or tablet computers, touch pads, multi-touch devices, smart phones, personal digital assistants, other multi-function devices, stand-alone kiosks, etc. These, or any combination of these, may have utility in designing the mechanical components of the systems illustrated in the drawing figures, as well as in computing forces, stresses, strains, and expected life spans of given structural members under a given set of assumptions and performance requirements.

[0038] Turning now to the drawing figures, FIG. 1A is a simplified side perspective view of a prior art reinforced concrete wall, and FIG. 1B is a simplified vertical cross-sectional view of the prior art wall of FIG. 1A, taken along a transverse axis (reference numeral 172) illustrated in FIG. 1A. By way of background, it is noted that a few fundamental engineering principles have historically influenced design approaches and construction methodologies of both reinforced masonry walls

and reinforced concrete walls. Traditional (and current) structural engineering design approaches related to these two common structural members are generally governed by strict adherence to these fundamental principles, some of which are illustrated in FIGS. **1A** and **1B**.

[0039] The design of any wall generally requires creating a structural member having adequate resistance to forces and vectors (where a vector comprises a force component and a directionality component). Particularly in the case of an exterior wall (i.e., one having an exterior surface exposed to an exterior of a structure, and thus being exposed to wind and other environmental elements), adequate resistance in this context generally requires simultaneous support in opposition of both vertical loads (i.e., gravity and uplift roof loads) as well as horizontal or lateral loads (e.g., wind loads). Since gravity loads (particularly in the context of residential construction projects) are typically very small relative to the maximum compression loads that even conventional masonry or concrete structures can support, it is instead the lateral loads that are expected to be exerted on a particular wall segment which generally control the specifications, design, and construction of that particular wall segment. As a result, many construction engineers and crews design and build the wall as (in essence) a vertically oriented beam that is subjected to wind load vectors which act generally perpendicular to the wall's plane in two principal horizontal directions, depending on the wind direction, namely: inward oriented vectors (e.g., when the exterior surface of the wall is on the windward side of the structure, and the wind is incident on the exterior surface), or outward oriented vectors (e.g., when the exterior surface of the wall is on the leeward side of the structure, and the lateral forces experienced by the wall are communicated through the structure, which is typically referred to as "internal pressure, and when wind shedding off a building causes suction or a partial vacuum, creating an outwardly directed vector on an exterior surface). Specifically, the wall must be designed to act as a beam capable of resisting forces applied from either direction, which is exactly how the prior art wall section (reference numeral **100**) of FIGS. **1A** and **1B** is designed and constructed.

[0040] As depicted in the drawing figures, prior art wall **100** has a top **101** and a bottom **102**, the distance between top **101** and bottom **102** generally defining a height of wall **100**. It will be appreciated that top **101** typically supports a floor joist (or a plurality of joists) serving as a base for an additional story, a roof sill or plurality of rafters to support a roof, or other structural members as is generally known in the art, and that bottom **102** is either supported by one or more floor joists, support beams, or a foundation (or a foundational slab), for example, all of which would be configured and operative to carry the weight of wall **100** and the other structural components supported by wall **100**. These ancillary elements are omitted from FIGS. **1A** and **1B** for clarity, but it is noted that the illustrations of wall **100** are intended to denote a typical wall having typical connections to other structural elements comprising a house, building, or other structure.

[0041] Wall **100** generally has an interior surface **110** (on an interior, protected, or leeward side) and an exterior surface **130** (on an exposed, weather, or windward side). These surfaces **110** and **130** are usually substantially parallel to each other at a uniform distance apart, which distance along transverse axis **172** defines a thickness ("T" in FIG. **1B**) of wall **100**, and span a horizontal distance substantially parallel to a longitudinal axis (reference numeral **171** in FIG. **1A**). In the case of masonry or concrete walls which are generally more capable under compression than they are under tension or torsion, a reinforcing system **190** may be provided in the internal structure of wall **100** to strengthen wall **100** against tensile stresses, horizontal forces, or oblique vectors that may cause bending or twisting as opposed to compression. In conventional arrangements, reinforcing system **190** may generally comprise steel bars or mesh oriented horizontally (reference numeral **191**) and vertically (reference numeral **192**), though other arrangements are possible. It is noted that the implementation of reinforcing system **190** is abstracted in FIGS. **1A** and **1B** for clarity, and that the number and position of horizontal components **191** and vertical components **192** may vary as a function of the design and purpose of wall **100**, the material, thickness, gauge, hardness, and ductile strength of the various components **191** and **192**, or a combination of these and a variety of other

factors.

[0042] Lateral wind forces (represented by the block arrows in FIG. 1B) incident on exterior surface **130** of wall **100** create flexural stresses in the form of a “moment” or bending force. In the case of a wall of the reinforced concrete or masonry variety (such as wall **100**), a universal moment resistance equation defines a moment resistance value for wall **100** as a direct function of a distance between the outer compressive layer of wall **100** (in this case, exterior surface **130**) and a center of the structural components **191** and **192** comprising reinforcing system **190** (as viewed in a vertical cross-section taken along transverse axis **172**). This distance (referred to as a “d” value in reinforced concrete moment resistance calculations) is illustrated at “d” in FIG. 1B, and is defined as $d=T/2$; i.e., reinforcing system **190** is conventionally disposed substantially equidistant from interior surface **110** and exterior surface **130**. The representations in FIGS. 1A and 1B are consistent with current prior art design and construction paradigms, which place reinforcing system **190** at a center of thickness T of wall **100** structure.

[0043] As noted above, such traditional placement of reinforcing system **190** relative to thickness T of wall **100** is generally due to the fact that wall **100** is intended to be configured and operative to resist lateral forces in two directions (i.e., both inward vectors and outward vectors); placing reinforcing system **190** at a center (along transverse axis **172**) of wall **100** such that $d=T/2$ establishes a substantially equal reinforcing functionality for both wind load vector directions. However, by placing the mass of components **191** and **192** of reinforcing system **190** midway between interior surface **110** and exterior surface **130**, by design, when a wind load vector is in the inward direction (e.g., from right to left in FIG. 1B), roughly half of the cross-sectional area of wall **100** (the left half in FIG. 1B) is providing no resistance to flexural stresses induced by the lateral load, and therefor is serving no structural purpose in that regard. When the wind load vector is in the other direction (e.g., from left to right in FIG. 1B), while that same cross-sectional area may fulfill its structural purpose in compression, then the opposite portion of wall **100** (the right half in FIG. 1B) becomes structurally ineffective to resist the new lateral load induced flexural stresses. Though some resistance to shear may be provided, in each situation noted above, a portion of the core of wall **100** is not effectively handling lateral loads, and thus wall **100** is “over-engineered” to compensate for its own inefficient design.

[0044] This is what is referred to as a “wasteful dilemma” of current wall design and construction; purposefully and by design, the “d” value (which is a direct representation of a moment resistance capacity) is based upon only one half of thickness T of a conventional wall **100**. Even the building code promulgated by the American Concrete Institute (specifically, ACI-318), which is the preeminent code for reinforced concrete construction in the United States, assumes uniform wall cross-sectional area in all relevant code provisions. The same is true for reinforced masonry construction design. Not only is the moment resistance capability of a uniformly thick wall having a reinforcing structure at its transverse midpoint limited considerably because the “d” value is limited to only half of the wall's thickness, but also, those of skill in the art will appreciate that almost half of the concrete or masonry utilized for construction of such walls is structurally non-functioning (at least vis-à-vis lateral loading), depending upon the wind (or other source of lateral load vector) direction.

[0045] As set forth in detail below, aspects of the present disclosure address the forgoing limitations inherent in prior art wall **100**, providing both a wall that has novel geometric characteristics to minimize wasted structural material and an efficient mechanical system and methodology to create such a wall. The walls, systems, and methods illustrated and described herein were developed to optimize use of construction materials that are necessary to achieve the wall's function and desired or required structural characteristics; additionally, the systems and methods to create such a wall were developed to minimize human labor and manhours required while still ensuring quality construction results. In many implementations, the use of interchangeable parts or components that are reusable and portable facilitates construction

processes that are efficient and repeatable, such that a construction crew may consistently create reinforced walls or other structures having predictable performance characteristics with minimal effort as compared to conventional construction methodologies.

[0046] In that regard, FIG. 1C is a simplified side perspective view of one implementation of a reinforced structural member according to one aspect of the disclosed subject matter, and FIG. 1D is a simplified vertical cross-sectional view of the reinforced structural member of FIG. 1C, taken along a transverse axis (reference numeral 272) illustrated in FIG. 1C. Additionally, FIG. 1E is a more detailed side perspective view (from the opposite side) of the reinforced structural member of FIG. 1C showing a span having a baseline thickness between adjacent spans having a thickness that is less than the baseline thickness; specifically, the perspective shown in FIG. 1E is from the interior side of reinforced structural member 200, whereas the perspective in FIG. 1C is from the exterior side. It is noted that FIGS. 1C through 1E are abstracted to facilitate comparison with the prior art wall 100 illustrated and described with reference to FIGS. 1A and 1B. As illustrated in the drawing figures, a structural member (reference numeral 200) may generally comprise many of the same elements as conventional wall 100, with some notable distinctions. In a departure from conventional paradigms, structural member 200 may generally include a recessed region (indicated by reference numeral 299) that eliminates some structural material. As best seen in FIGS. 1D and 1E, recessed region 299 may be formed in a portion of interior surface (reference numeral 110) and extend toward exterior surface (reference numeral 130) such that a recess surface 297 is closer to exterior surface 130 than interior surface 110 is; as depicted in the drawing figures, in some implementations, recessed region 299 may be characterized by a recess height that is less than the height of structural member 200, a recess width that spans a horizontal distance of interior surface 110 (e.g., along a longitudinal axis 271), and a recess depth (reference numeral 298) from interior surface 110 that is less than the baseline thickness “T” of structural member 200. In practice, recessed region 299 generally defines an effective thickness (“T.sub.e” in FIG. 1D) of structural member 200 that is equal to a difference between the baseline thickness T and recess depth 298 (i.e., from exterior surface 130 to recess surface 297).

[0047] During construction of structural member 200, portions of a reinforcing system 700 may be offset from a centerline between interior surface 110 and exterior surface 130 and biased towards exterior surface 130. As set forth in more detail below, a wall plane reinforcement element may be embodied in or comprise rebar, other metal structures, wire or steel mesh, or fabric or other fibrous materials such as are generally known in the art (reference numeral 791). In some implementations, wall reinforcement element 791 is disposed intermediate recess surface 297 (i.e., at recessed region 299) and exterior surface 130, supports a portion of structural member 200 having effective thickness T.sub.e, and is selectively positioned relative to exterior surface 130 such that a resistance to a given bending moment for structural member 200 as a whole is optimized for the amount of construction materials used. In some implementations, an additional columnar or post component (“post reinforcement element,” reference numeral 793) of reinforcing system 700 may be deployed at areas of wall 200 which do not include recess region 299; such a post reinforcement element 793 is discussed in more depth below with specific reference to FIG. 7B. It will be appreciated that post reinforcement element 793 may be embodied in or comprise steel rebar, metal pipe, cable, or other elongate reinforcing material such as is common in the construction industry. Additionally, in some implementations discussed below, reinforcing system 700 may further comprise a beam reinforcement element (reference numeral 772, not shown in FIGS. 1C through 1E) that is configured and operative to provide reinforcement to a portion of reinforced structure 200 above fenestrations such as windows and door openings where a horizontal beam component is used to transfer loads around the opening.

[0048] Those of skill in the art will appreciate that the description of FIGS. 1C through 9 sets forth an extremely economical and time efficient system and method of constructing reinforced walls or other structural members that are structurally sound (e.g., capable of safely resisting all normal

axial or vertical loads as well as violent lateral loads), precisely arranged, and consistent from one wall or structure to the next. In this context, the term “wall” is intended to be broad enough to include other load-bearing structural components such as wall segments, columns, arches, and the like. In some instances, a wall such as set forth below may be capable of resisting lateral loads exerted by winds from hurricanes of category 5 (i.e., on the Saffir-Simpson Hurricane Wind Scale used by the National Hurricane Center, a branch of the National Oceanic and Atmospheric Administration) or tornadoes of category EF3 (i.e., on the Enhanced Fujita Scale); these are sustained winds approximating, or even exceeding, 160 miles per hour. The walls such as reinforced structural member **200** described herein may be made to be geometrically accurate in all three dimensional parameters, may have a useful life in excess of 100 years while requiring minimal to virtually no maintenance, and may be incombustible, water resistant, and constructed for approximately half the cost of a conventional wall system having similar strength and functional characteristics.

[0049] Turning now to systems and methods of constructing such a wall or other building component, it is noted that FIGS. 2A through 2E show various elevation, side, and perspective views of a base plate and a brace shoe for use in connection with a system for forming a reinforced structural member. Specifically, FIG. 2A is a side view of a base plate **220** disposed upon and secured to a foundation **290**. In FIG. 2A, foundation **290** is illustrated as supported by earth (reference numeral **295**), which is typically prepared in some manner, such as by leveling and compacting, for instance, and may also be roughed or covered with gravel or other materials, to support foundation **290** as a base for a building or other structure. It is noted, however, that the term “foundation” in this context (and as represented at reference numeral **290**) is intended to be broad enough to include a floor structure of a lower story, such as a joist or a joist hanger, a sill plate, or a beam, for example, or other physical component of a building or structure. In use, any surface or structural component that is configured and operative to support the weight of a reinforced structural member **200** may be suitable for foundation **290**.

[0050] During construction of reinforced structural member **200**, a base plate **220** and, optionally, a brace shoe **240** (see FIGS. 2C through 2E), may be selectively (and removably) secured to foundation **290**. This may be accomplished with bolts, screws, concrete or other anchors, or other elongate fastening members (not shown in the drawing figures for clarity) suitably selected as a function of the materials and design characteristics of foundation **290**, for example, and the materials selected for base plate **220** and brace shoe **240** as described below. In some implementations, base plate **220** and brace shoe **240** may comprise apertures **229** and **249**, respectively, such as pre-drilled holes to accommodate such fastening members and to facilitate selectively securing base plate **220** and brace shoe **240** to foundation **290**. Additionally or alternatively, base plate **220**, brace shoe **240**, or both, may be selectively secured to foundation **290** using adhesives, epoxy, resins, or other substances suitable for making mechanical connections, depending upon the materials used for foundation **290**, base plate **220**, and brace shoe **240**.

[0051] In that regard, base plate **220**, brace shoe **240**, or both, may be constructed of metal (such as steel, stainless steel, aluminum, titanium, bronze or other alloys, and the like), ceramic, glass, fiberglass, carbon composites, or other materials generally known in the construction arts. The present disclosure is not intended to be limited by the particular materials used for base plate **220** or brace shoe **240**, though it is noted that the specific dimensions of these components may vary as a function of the materials employed, and that the materials should be selected to have strength, weight, and corrosion resistance characteristics that are suitable for an intended use and an expected life-span of the components.

[0052] As illustrated, base plate **220** may be elongate and comprise a generally horizontal plate portion **227** which may be selectively attached or secured to foundation **290** in an orientation that is substantially perpendicular to a longitudinal axis of reinforced structural member **200** (i.e., perpendicular to longitudinal axis **271** as best depicted in FIGS. 2D and 2E) at a point that is

interior to a footprint (reference numeral **291**) where a completed reinforced structural member **200** will eventually stand. Base plate **220** further comprises a column support bracket **221** disposed at or near an end that is proximate to footprint **291**, a brace member support flange **222** disposed at or near an end that is distal to footprint **291**, and, optionally, one or more apertures **229** to facilitate securing base plate **220** to foundation **290** as set forth above.

[0053] Similarly, brace shoe **240** may be square, circular, polygonal, or elongate (though appreciably shorter than elongate base plate **220**) and comprise a generally horizontal plate portion **247**, a brace member support flange **242** disposed at or near an end that is positioned proximal to column support bracket **221** and, optionally, an aperture **249** to facilitate securing brace shoe **240** to foundation **290** as set forth above. In use, brace shoe **240** may be selectively attached or secured to foundation **290** in an orientation that aligns brace member support flange **242** substantially parallel to longitudinal axis **271** (as best depicted in FIGS. 2D and 2E) at a point that is interior to footprint **291**, and generally aligned laterally with column support bracket **221**.

[0054] By way of background, a wall such as reinforced structural member **200** may be conceptualized fundamentally as the connection of four corner points located in three-dimensional space by a rigid plane. Given four corner points located precisely in space, then a rigid plane having corners located exactly at these four points will establish a surface of the structural member (such as **200**). Mirror this surface about a longitudinal axis (such as **271**), and the other, opposite surface of the structural member **200** is established. From the perspective of a construction company or site crew, one engineering objective is to be able to accomplish this with precision and efficiency, with minimal time and effort expended by skilled laborers.

[0055] With this context, it will be appreciated that base plate **220** serves two functions: two adjacent base plates **220** may be used as a rigid point of connection as set forth in detail below to establish two out of the four corner points (the two bottom points) of the surface of the structural member **200**; and each base plate **220** further creates a rigid connection point for lateral bracing of vertical components of a construction system framework to be described below. This is done with a single part (horizontal plate portion **227**) having two fasteners (column support bracket **221** and brace member support flange **222**) selectively connected to the supporting foundation **290** (such as a concrete slab, floor joist, sill, beam, or the like). Complementing operation of base plate **220** in use, brace shoe **240** may generally create a rigid connection point for lateral bracing of vertical components in the opposite principal direction (i.e., substantially perpendicular) from base plate **220**. In some implementations, one or more brace shoes **240** may be optimally located at or near a building's or other structure's primary outer corners.

[0056] As noted above, specific materials and dimensions of base plate **220** and brace shoe **240** may vary as application-specific design choices, but it will be appreciated that they may be so designed and implemented safely to support all forces and vectors to which they are intended to be subjected throughout a given application (i.e., a given construction process).

[0057] FIGS. 3A and 3B show, respectively, a side view and a plan view of a column and brace members for use in connection with a system for forming a reinforced structural member. Specifically, as part of erecting such a system for forming a reinforced structural member **200**, a vertical system assembly **300** may generally comprise a column **320** suitably attached to column support bracket **221** of base plate **220** and braced at brace member support flange **222** of base plate **220** with one or more bracing members **340** substantially as illustrated in FIGS. 3A and 3B.

[0058] As with base plate **220** and brace shoe **240**, column **320** and bracing member **340** may be constructed of any of various suitable metals, ceramic, glass, fiberglass, carbon composites, or other materials generally known in the art, and are not limited to any particular material, though it will be appreciated that the specific dimensions (including diameter, thickness, wall thickness in the case of hollow embodiments, etc.) of these components may vary as a function of the materials selected and their respective strength and ductility characteristics.

[0059] As best illustrated in FIG. 3A, column **320** may generally be embodied in or comprise an

elongate cylinder or tube having a height selected to accommodate a height of the structural member (such as reference numeral **200**) to be fabricated, accounting for additional system components described below. Column **320** may be generally configured and operative to cooperate with column support bracket **221** of base plate **220**. For example, column **320** may include a recessed or “female” lower portion that is designed and operative to engage a “male” protuberance of column support bracket **221** such that column **320** fits snugly on column support bracket **221**. In the case where column support bracket **221** is circular in plan (e.g., see FIG. 2B), then column **320** may likewise be circular in planar cross-section, though other implementations are contemplated, such as instances in which both column **320** and column support bracket **221** have matching plan shapes that are not circular (such as oval, square, rectangular, or other polygonal shapes). As another example, column **320** may generally be of square or polygonal planar cross-section, but include a circular female coupling to engage a circular column support bracket **221**, or vice versa. In cases in which column **320** is implemented as a hollow tube (of any desired planar cross section), such a female coupling may not be necessary, as the hollow structure of column **320** may be so sized and dimensioned as to engage column support bracket **221** directly, such as via slip-fit or friction-fit. In any event, column **320** is to be engaged with column support bracket **221** (and perhaps secured, for example, with set screws or other mechanical fastening elements) and braced in a substantially vertical orientation as illustrated in FIG. 3A.

[0060] To accommodate lateral support during use, column **320** may also include a brace member attachment element generally identified by reference numeral **350** in FIG. 3A. Brace member attachment element **350** may be embodied in or comprise a flange, bracket, or aperture such as a pre-drilled hole for accepting or cooperating with a complementary structure of brace member **340**, the distal end of which may be mechanically coupled to brace member support flange **222** of base plate **220** (see FIG. 3A) or, optionally, brace member support flange **242** of brace shoe **240** (see FIG. 3B). In that regard, brace member attachment element **350** may include two attachment points oriented at 90 degrees such that two orthogonal brace members **340** may be attached to provide support to column **320**, respectively, in one of two primary force directions (i.e., along longitudinal axis **271** and transverse axis **272**) as depicted in FIG. 3B. Additionally or alternatively, column **320** may include more than one brace member attachment element **350**, spaced vertically, for example, to accommodate bolts, wing nuts, thumb screws, set screws, or other fastening components that may be desired or required to establish rigid connection with more than one brace member **340**. The present disclosure is not intended to be limited by the type of mechanical connection used to effectuate physical coupling of column **320** and brace member **340** at brace member attachment element **350**, though it is noted that such a connection may be sufficiently adjustable to enable support of column **320** in a substantially vertical, or plumb, orientation. In some such implementations, brace member **340** may be made to telescope, for instance, with cooperating threaded portions or turnbuckles, such that an overall length of brace member **340** (from base plate **220** or brace shoe **240** to column **320**) may be selectively adjusted to maintain column **320** in a substantially vertical orientation. In this context, the term “substantially” vertical or plumb will be appreciated by those of skill in the art to mean that column **320** is supported close enough to vertical or plumb to satisfy applicable building codes, industry standards, or specific job-site requirements.

[0061] Brace member **340** may comprise two structures (such as flanges, tabs, grooves, apertures or pre-drilled holes, snap-together fasteners, or the like) each of which is disposed at a respective end of brace member **340**, and which are respectively configured and operative to cooperate with or otherwise to engage either brace member attachment element **350** or brace member support flanges **222** and **242**, substantially as illustrated in FIGS. 3A and 3B. As noted above, it may be desirable that a length of brace member **340** may be made to be selectively adjustable, such as on a job site. This may be accomplished with cooperating threaded portions or turnbuckles that allow relative longitudinal movement of rigid portions of brace member **340**.

[0062] It is noted that in the illustrated implementation, column **320** is designed to mate with column support bracket **221** using a slip-fit engagement, enabling a quick and easy installation requiring only minutes of labor. In this circumstance, it may be desirable to include apertures or pre-drilled holes in the cooperating structures (of one or both of column **320** and column support bracket **221**) to accommodate fasteners (such as bolts or set screws) to ensure proper placement and orientation of column **320**, for instance, to ensure that one or more brace member attachment elements **350** are properly aligned to couple with one or more brace members **340**. In the case of a column **320** having a square or rectangular planar cross-section and a column support bracket **221** having a circular planar cross-section (as in the illustrated arrangement), such fasteners may have utility in preventing a work crew from installing column **320** on column support bracket **221** in any orientation or alignment other than as intended.

[0063] When one or more brace members **340** are installed as illustrated (and their respective lengths optionally adjusted as set forth above), column **320** may then be supported substantially plumb relative to the supporting surface (such as foundation **290**), even in the event of minor elevation changes or imperfections in the supporting surface (an inevitable outcome). This entire process may be accomplished in a matter of minutes for each column **320** to be deployed. After properly braced and plumbed, the top portion of two adjacent columns **320** may then be used as set forth in detail below to establish the remaining two points (i.e., the top two points) defining principal corners of a portion of reinforced structural member **200** as described above.

[0064] In some implementations, column **320** may also comprise a top support attachment point **371** and a lower rail attachment point **372**. These attachment points **371** and **372** may be constructed to engage with cooperating structures on a top support structure and a lower rail assembly described below. Attachment points **371** and **372** may generally be embodied in or comprise apertures such as pre-drilled holes or slots, for example, or brackets, clips, or flanges integrated with structure of column **320**, depending upon the structural characteristics and implementation details of the top support structure and the lower rail assembly, respectively, with which they are designed to couple, the materials from which column **320** is constructed and its planar cross-section, or a combination of these and a variety of other factors. The use and purpose of these attachment points **371** and **372** will become apparent in the discussion of the other system framework components set forth below.

[0065] As noted above with reference to other physical elements of a system for fabricating a reinforced structural member, the specific materials and dimensions of column **320** and brace member **340** may vary as application-specific design choices, but it will be appreciated that they may be so designed and implemented safely to support all forces and vectors to which they are intended to be subjected throughout a given application (i.e., a given construction process).

[0066] FIGS. **4A** through **4C** show a side view of a top support structure, and plan views of two implementations of a top support structure, for use in connection with a system for forming a reinforced structural member. For most situations, a top support structure **400** may generally comprise a vertical member **470**, a horizontal portion **420**, and a column attachment point **450**. In use, vertical member **470** of top support structure **400** may slip-fit over an outside perimeter of column **320** particularly in instances in which column **320** is tapered near the top and adapted to accommodate it; alternatively, vertical member **470** may be so dimensioned to slip-fit within an inside volume of column **320**, particularly in instances in which column **320** is hollow near the top and adapted to accept it. In some implementations, cooperating threads may be used to facilitate engagement between vertical member **470** and column **320**, although it is noted that proper alignment of horizontal portion **420** may be challenging in this arrangement (one or more detent mechanisms or careful machining of the cooperating threads may have utility in ensuring that a fully threaded engagement results in proper alignment of horizontal portion **420** during use). In the drawing figures, vertical member **470** is dimensioned to slip-fit snugly within a cavity near the top of column **320**, and includes a flange or other elongate structure at column attachment point

(reference numeral **450**) that is sized, dimensioned, and oriented to engage a cooperating slot or other aperture at support attachment point **371** (see FIGS. **4B** and **4C**). This arrangement may ensure proper alignment of horizontal portions **421** and **422** as described below.

[0067] The primary purpose of top support structure **400** is to position horizontal portion **420** (and in some instances, more than one such horizontal portion, as indicated at reference numerals **421** and **422** in FIG. **4C**) over a footprint of the structural member to be fabricated (such as footprint **291** in FIGS. **2D** and **2E**). Specifically, horizontal portion **420** may include one cantilevered support **421** (FIG. **4B**) positioned in a substantially horizontal orientation and substantially parallel to transverse axis **272**; cantilevered support **421** may have a length selected such that it extends over the footprint **291** as indicated in FIG. **4A** sufficiently to accommodate other system components as set forth below. In some embodiments, horizontal portion **420** may include an additional cantilevered support **422** (FIG. **4C**) positioned in a substantially horizontal orientation and substantially perpendicular to the first cantilevered support **421**; in this configuration, which may have particular utility at a corner of a building where two orthogonal structures meet at a right angle, additional cantilevered support **422** may be so dimensioned as to extend beyond the footprint of the orthogonal structure. Angles other than right angles are contemplated, and may be facilitated as necessary or desired by adjusting an angle between cantilevered portions **421** and **422**. As noted above with reference to a vertical orientation, in this context, the terms “substantially” horizontal and “substantially” parallel will be appreciated by those of skill in the art to mean sufficiently horizontal or parallel, as the case may be, to comply with applicable building codes, industry standards, or job-specific requirements.

[0068] In the illustrated versions of FIGS. **4A** through **4C** and as noted briefly above, vertical member **470** is sized and dimensioned to slide into an appropriately dimensioned cavity disposed at a top portion of column **320**, and a flange (reference numeral **450**) engages a cooperating slot at top structure attachment point **371** of column **320**; in this manner, proper orientation of horizontal portion **420** may be ensured when vertical member **470** is seated properly on column **320** and flange **450** engages attachment point **371**. Other engagement structures are possible, including apertures or pre-drilled holes to accommodate bolts or screws, for example, and may be selected as a design choice, or may be dictated by the nature of the structure **200** to be fabricated and desired angles, if any, between cantilevered supports **421** and **422**, among other factors.

[0069] As noted generally above, FIGS. **4B** and **4C** are plan views of two different types of top support structure **400**, respectively: an inner top support having a single cantilevered support **421** for use in straight wall runs; and a corner top support having two cantilevered supports **421** and **422** disposed at right angles to each other, for use at a corner of reinforced structure **200** (as noted above, the angle between cantilevered supports **421** and **422** need not be right, and may be obtuse or acute, for instance, as structural or building plans require). Cantilevered supports **421** and **422** may include pre-drilled holes, for instance, or alignment indicia identifying proper attachment sites (reference numerals **423** and **424**, respectively) where other components supported by cantilevered supports **421** and **422** are to be attached. The specific number and location of attachment sites **423** and **424** may vary by application and materials used for cantilevered supports **421** and **422**, the weight they are intended to bear, or a combination of these and other factors. Additionally or alternatively, these components **421** and **422** may include vertical ridges or guides (e.g., “bumpers,” generally indicated at reference numeral **429**) to facilitate engagement of other system components carried by cantilevered supports **421** and **422**. In that regard, when a work crew member places a rail assembly (described below) on top of a cantilevered support **421** or **422** and attempts to reposition it for proper attachment at attachment sites **423** or **424**, as appropriate, bumpers **429**, which are raised slightly above the top surface of cantilevered supports **421** and **422** (see FIG. **4A**), will prevent the rail assembly from slipping off during installation.

[0070] As noted above with reference to other physical elements of a system for fabricating a reinforced structural member, the specific materials (such as, for example, steel, other metals or

alloys, glass or ceramics, fiber or other composites, and the like) and dimensions of top support structure **400** (specifically vertical member **470** and cantilevered supports **421** and **422**) may vary as application-specific design choices, but it will be appreciated that they may be so designed and implemented safely to support all forces and vectors to which they are intended to be subjected throughout a given application (i.e., a given construction process) without unwanted deflection or warping.

[0071] FIGS. 5A through 5C show a side view and plan views of a rail assembly attached to the top support of FIGS. 4A through 4C and operative for use in connection with a system for forming a reinforced structural member. As illustrated, rail assembly **500** may generally comprise longitudinal top chords **521** and **522**, tapered cross members **550** (one implementation of a taper is best illustrated at reference numeral **570** in FIG. 5A), and interior and exterior bottom or hanging chords (reference numerals **551** and **552**, respectively). From a structural perspective, rail assembly **500** may be constructed to function in much the same manner as a three-dimensional space frame, box beam, or truss with top chords **521**, **522** and hanging chords **551**, **552** being integrated with or otherwise fixedly attached to respective structural points of cross members **550**. As will be appreciated by those of skill in the art, rail assembly **500** may therefore exhibit significantly more resistance to bending and twisting than longitudinal stringers or chords operating independently.

[0072] In use, rail assembly **500** may be configured and operative to suspend the form structures and reinforcing system **700** (described below) that may be used to define reinforced structural member **200** to be fabricated; it may also support a movable dispensing component that may translate linearly along rail assembly **500** selectively to deposit construction materials (e.g., a material slurry) serving as the composition of reinforced structural member **200**. Where structural members **200** meet at a corner of a structure (e.g., at a right angle or otherwise), one rail assembly **500** for one structural member **200** may be positioned adjacent to and abutting another rail assembly **500** for the other structural member **200** as best depicted in FIG. 5C. Because the various components defining the reinforced structure to be fabricated are suspended from interior and exterior hanging chords **551** and **552**, the height of rail assembly **500** above foundation **290** may be selectively varied without impacting construction of reinforced structural member **200** or its height; specifically, one or more lengths of hardware elements used to suspend the other system components may be selectively varied to accommodate varying heights (with respect to foundation **290**) of multiple rail assemblies **500**, irregularities in foundation **290**, or both.

[0073] As best illustrated in FIGS. 5B and 5C, cross members **550** may be spaced longitudinally and rigidly join top chords **521**, **522** and hanging chords **551**, **552** at regular intervals. Cross members **550** may have apertures, pre-drilled holes, or alignment indicia as indicated in FIG. 5A to facilitate suspension of elements of reinforcing system **700** as set forth in detail below with reference to FIGS. 7A and 7B. The number and particular locations of these holes or indicia may be application-specific, and may, accordingly, vary as a function of the materials used for cross members **550** and their spacing, as well as the expected weight and position of the reinforcing system elements to be suspended, among other factors. In the FIG. 5A implementation (and as depicted throughout the drawing figures), holes or hanging locations are identified by an interior column **583**, a center column **582**, and an exterior column **581**. This arrangement has utility to the extent that hanging hardware suspended from both holes or attachment sites in a given column (**581**, **582**, or **583**, as the case may be) will be substantially plumb, and will remain so during use, whereas if only one hole or attachment site were used, additional care (and testing) would be required to ensure a proper vertical orientation of the component hung or otherwise supported therefrom.

[0074] It will also be appreciated that the number and spacing of cross members **550** themselves may vary by application, but in some instances, it may be desirable to keep the number and spacing uniform for a given longitudinal span; in this manner, any given rail assembly **500** may be interchangeable with any other on a job site, minimizing effort and risk of mistake on the part of a

job site work crew. Top chords **521** and **522** may include apertures, pre-drilled holes, or alignment indicia (reference numerals **531** and **532**, respectively) to facilitate proper engagement with, and orientation with respect to, cooperating elements or pre-drilled holes at appropriate attachment sites (see reference numerals **423** and **424** in FIGS. **4B** and **4C**) on cantilevered supports **421** and **422** to which top chords **521** and **522** are to be attached. It is noted that the specific number and precise locations of these holes or indicia (**423**, **424**, **531**, and **532**) may be application-specific, and may vary according to the materials used for, and the overall dimensions of, the various components, stiffness or rigidity requirements, and allowable tolerances per applicable code, among other factors. In that regard, it will be appreciated that the arrangement illustrated in the drawing figures is provided by way of example only, and that more or fewer points of attachment for top chords **521** and **522** and cantilevered supports **421** and **422**, and their locations, may be employed. Alternatively, it may be desirable in some circumstances that top chords **521** and **522** and cantilevered supports **421** and **422** are coupled, joined, or otherwise rigidly connected with adhesive, epoxy, or even via welding; again, this may be application-specific and a function of, for example, desired rigidity of the structural framework, weight of the system components to be suspended by the framework (and the distribution of that weight), code compliance criteria, or a combination of these and other factors.

[0075] In the illustrated implementation, top chords **521** and **522** may be securely attached (at points **531** and **532**, respectively) to cantilevered supports **421** and **422** (at points **423** or **424**, as applicable), for example, with bolts, screws, set screws, wing nuts, or other fasteners that are generally well known in the art, though other options are contemplated. In some instances, hardware employed to make this mechanical connection may be selected to have a “low profile” (e.g., in the side view of FIG. **5A**) to avoid interfering with operation of equipment deployed on top of top chords **521** and **522** (see, e.g., FIG. **8B**). For example, bolt heads and nuts may be counter-sunk into the structure of top chords **521** and **522** or cantilevered supports **421** and **422**; in the FIG. **5A** implementation, bolts are shown in dashed lines as connecting top chords **521** and **522** to cantilevered support **420**, with bolt nuts shown on the underside of cantilevered support **421** to allow for unimpeded operation of components on the top side of rail assembly **500**.

[0076] In use, rail assembly **500** may be constructed as a three-dimensional space frame or box truss that is structurally sized, dimensioned, and constructed simultaneously to resist all superimposed vertical gravity loads and lateral loads while exhibiting negligible deflection in both principal directions (i.e., along longitudinal axis **271** and transverse axis **272**), as well as negligible twisting or torsional deformation. Accordingly, the number, spacing, and thickness of cross members **550** per unit length in the longitudinal direction, as well as the depth and profile of taper **570**, may be selected as a function of the overall length of rail assembly **500** and the weight of the construction components to be supported, among other factors such as the material used and its structural characteristics and performance parameters (such as hardness, rigidity, temperature tolerances, and the like).

[0077] Those of skill in the art will appreciate that rail assembly **500** configured and constructed as set forth herein represents an innovative support platform for both suspending wall forms (such as at bottom chords of the box truss represented by hanging chords **551** and **552**) and a track system (such as at top chords of the box truss represented by top chords **521** and **522**) for additional equipment such as a deposition hopper or trolley to deposit construction materials as set forth below with reference to FIG. **8B**. As described above with reference to FIGS. **2D** through **5C**, and as best illustrated in FIG. **8A**, components of a system of fabricating a reinforced structural member are arranged such that rail assembly **500** is purposely located above and spanning a geometry of the structural member **200** to be fabricated. Further, the geometric design of rail assembly **500** allows for efficient installation of wall forms (as described below) by creating a rigid, straight, and level line that may be used to support the forms that establish the two final principal points of structural plane described above (i.e., the top two points). At this point, it will be

apparent to those of skill in the art that rail assembly **500** is operative to support all wall system components (i.e., forms and reinforcing system elements) by suspending them from the top down, rather than by building them from the bottom up. This eliminates any need for supporting these elements directly from the ground (reference numeral **295**) or foundation (reference numeral **290**); accordingly, any undulations, slopes, or other imperfections associated with the underlying ground **295** or foundation **290** do not affect the wall or other structure **200** to be fabricated. In addition, reinforcing steel may be pre-fabricated welded wire mesh sheets that may be hung from rail assembly **500** in the same manner as forms, facilitating efficient installation and placement of components of reinforcing system **700** within an inner core of the structure **200** to be fabricated, as described in more detail below with reference to FIGS. 7A, 7B, and 8A.

[0078] As noted above with reference to other physical elements of a system for fabricating a reinforced structural member, the specific materials (such as, for example, steel, other metals or alloys, glass or ceramics, fiber or other composites, and the like) and dimensions of rail assembly **500** (as well as its constituent components) may vary as application-specific design choices, but it will be appreciated that they may be so designed and implemented safely to support all forces and vectors to which they are intended to be subjected throughout a given application (i.e., a given construction process) without unwanted deflection or twisting.

[0079] FIGS. 6A, 6B, and 6D through 6I show various side and plan views of implementations of a form selectively suspended from the rail assembly of FIGS. 5A through 5C and operative to define an interior surface of a reinforced structural member.

[0080] Initially, it is noted that an interior wall form **600** may generally be embodied in or comprise a conventional wall form such as is typically employed in the construction arts. Those of skill in the art will appreciate that the utility of the disclosed systems and methods is not intended to be limited by the nature, size, shape, and characteristics of interior wall form **600**.

[0081] With that in mind, however, it is also noted that, in some implementations, form **600** may be constructed with a frame fabricated of high strength structural aluminum (or other metal or alloy) or composite material and having framing components or support structures that are precisely placed to maximize strength, minimize deflections, and reduce weight of overall form **600**, facilitating installation of same by suspension from appropriate elements of rail assembly **500** as set forth above. In some implementations of such a reinforced or braced form **600**, a primary surface or “skin” may be fabricated of ultra-high strength, lightweight fiberglass (or other composite material) having a “T” beam vertical cross-sectional profile (in which the bottom of the “T,” the web of the beam, extends away from the exterior surface of the form as a ridge or elongate protrusion) to provide rigidity and lateral support. This skin may be attached to the high strength metal frame defining the overall shape of form **600** using techniques that are generally known in the art.

[0082] In practice, a number of vertically oriented, elongate “T” beams may be strategically placed to run at horizontal intervals which may be uniform or varied, for instance, as a function of the location and spacing of cross members or other framing components that support the skin, and perhaps relative to the overall height of the structural member **200** to be constructed, expected load distribution, the material properties of the materials used for form **600**, or a combination of these and a variety of other factors. The resulting profile may create a structural skin for form **600** that may exhibit a high moment of inertia (resistance to bending) due to the protruding miniature “beam webs” and the ultra-high strength fiberglass or other material employed. For example, some fiberglass or other composites may exhibit resistance to tensile loads of up to or even exceeding 55,000 pounds per square inch (stronger than some commonly used structural steel compositions). Such a construction approach may allow for a lightweight, yet strong and deformation resistant, panel skin for form **600**, and in some cases, the use of a “T” beam strategy for the skin profile may compensate for the inherently low modulus of elasticity (resistance to strain) of fiberglass or other lightweight composite materials used in construction of form **600**. When the composite skin is

applied or otherwise attached to the frame cross members, the resulting form **600** may be configured and so dimensioned such that each frame component and the overall skin surface is structurally loaded to capacity and located to create substantially uniform and negligible skin deflection (resulting in a uniformly fabricated structural member **200**), even though form **600** may be required to withstand a triangular load distribution (up to about 1,400 pounds per square foot for a 10 foot high wall, perhaps higher for a higher wall) when wet construction material slurry is poured.

[0083] Returning now to the drawing figures, FIG. **6H** is an interior side view, and FIG. **6I** is a top cross-sectional view, illustrating construction details of one implementation of an interior recess form **601**. In this embodiment, form frame **650** may generally comprise vertical members **652** and horizontal members or cross beams **630** that are fixedly attached to vertical members **652** creating a three-dimensional frame (see FIG. **6I**) that has sufficient strength and rigidity to withstand forces applied during use. A form skin **659**, which will form baseline surface **611** and recess surface **621** of form **601**, may be supported by beam webs **657** at horizontal cross beams **630** as illustrated. As noted above, vertical members **652** and cross beams **630** of frame **650** may be fabricated of high strength structural steel or aluminum (or other metal or alloy) or composite material, and may be dimensioned and spaced (note the varying vertical distribution in FIG. **6H**) in accordance with expected loads to be encountered during use. As noted briefly above, skin **659** and beam webs **657** may be fabricated of lightweight fiberglass or other composite materials (or even metals or alloys) having suitable strength and ductility characteristics for an intended application.

[0084] FIG. **6A** is a side view, and FIG. **6B** is a top view, of one implementation of an interior recess wall form **601** suspended from rail assembly **500**. Specifically, since recess form **601** is an interior form, it is illustrated as being suspended from interior hanging chord **551** coupled to cross member **550** of rail assembly **500**. Further, the overall skin **659** of recess form **601** includes a region that is protruding or projecting in plan such that use of recess form **601** during construction facilitates creation of the reinforced structural member **200** having recess region **299** as described above with reference to FIGS. **1C** and **1D**.

[0085] Recess form **601** is illustrated at having a baseline surface **611** to define an interior surface, and thereby the baseline thickness, of reinforced structural member **200**, as well as a recess element **620** that slopes to a recess surface **621**, which defines a recess depth (reference numeral **298** in FIG. **1D**) of a recessed region (reference numeral **299** in FIGS. **1C** and **1D**) that is to be formed in the resulting reinforced structural member **200**. This is best illustrated in the plan view of FIG. **6B**, which depicts a plurality of recess forms **601** adjacent to each other and each selectively suspended from interior hanging chords **551**. In this view (see also the plan view of FIG. **6I**), it can be seen that recess surface **621** is projected away from baseline surface **611** such that recessed regions **299** will be devoid of building material when reinforced structural member **200** is complete.

[0086] It is noted that a given recessed region **299** (having a recess depth **298**) may span an horizontal portion of recess form **601** that is less than the overall horizontal width (i.e., along longitudinal axis **271**) of recess form **601** itself. By abutting two or more such recess forms **601** next to each other (FIG. **6B**), a structural member **200** having an arbitrary horizontal length may be created having regular recessed regions **299** spaced in accordance with the shape of the particular recess forms **601** used, and whether one or more baseline forms (described below) are interspersed between successive recess forms **601**. As noted above in connection with FIGS. **1C** and **1D**, areas of a fabricated structural member **200** at recessed regions **299** have an effective thickness ($T_{sub.e}$ in FIG. **1D**) that is less than the baseline thickness, and which may be measured as the distance between recess surface **621** and an opposing surface (see reference numeral **811** in FIG. **8A**) of an exterior wall form as described below with reference to FIG. **8A**. It is noted that the horizontal spans between recessed regions **299** are of the baseline thickness, and that it is at these areas of recess form **601** where through-bores or suitably dimensioned slots between adjacent forms (reference numeral **690** in FIG. **6A**) may be disposed in recess form **601**. In some implementations,

such slots **690** may be located at edge bumpers (reference numeral **640** in FIGS. **6A** and **6I**) where two adjacent forms **600** abut each other. Bumpers **640** may be fabricated of rubber, silicone, plastic, or other pliable material such that vertical members **652** of adjacent form frames **650** are not damaged when two forms are placed side-by-side during installation. In use, slots **690** allow for insertion of tension ties which prevent relative movement between recess form **601** and an adjacent interior form **600** and between recess form **601** and an opposing exterior form during use as described in detail below with specific reference to FIG. **8A**.

[0087] The shape of recess form **601**, and in particular, the shape and contours of recess element **620**, are susceptible of numerous variations. For example, recess element **620** may be oval, oblong, rounded, square, or rectangular in plan, cross-section, or both; alternatively, in the trapezoidal implementation illustrated, the slopes or angles may vary as desired for a particular application, taking care to adjust locations of beam webs **657**, if applicable, as necessary or desired to ensure proper engagement with cross beams **630** of form frame **650**. As another alternative, the vertical height, horizontal span, or both, of a given recess element **620** may be modified as a function of specific design constraints or performance requirements for the reinforced structural member **200** to be fabricated. For example, a single recess form **601** may comprise more than one recess element **620** distributed horizontally along longitudinal axis **271** in some implementations; similarly, such a single recess form **601** may comprise more than one such recess element **620** distributed vertically.

[0088] FIG. **6A** also illustrates a plurality of cross beams **630**, distributed vertically along recess form **601** (see also the interior side view of form frame **650** in FIG. **6H**). In this implementation, cross beams **630** are elongate structural elements of form frame **650**, running horizontally along longitudinal axis **271**, to provide support in resisting forces applied to form **601** by skin **659**. As is evident in the drawing figures, vertical distribution of cross beams **630** may be more dense at a lower portion of recess form **601** (where forces will be greatest during use) than at an upper portion of recess form **601** (where forces will be relatively less). The specific number and vertical distribution of cross beams **630** may be application-specific as noted above, and may vary as desired to satisfy particular requirements or design preferences; the arrangement of FIGS. **6A** and **6H** is provided by way of example only.

[0089] Also in connection with resisting forces during use, it is noted that FIG. **6C** is a plan view of a lower rail assembly operative to secure a form (such as recess form **601**) to a lower portion of a support column (such as column **320**) as best illustrated in FIGS. **6A** and **6E**. Lower rail assembly **670** may be physically attached to lower rail attachment point **372** of column **320** as noted above in connection with FIG. **3A**. Specifically, each respective column connector **677** may be designed to engage a respective lower rail attachment point **372** on adjacent columns **320**, such that lower rail **671** spans (and rigidly connects) two adjacent columns **320**. Cooperating structures having utility for column connector **677** and lower rail attachment point **372** are generally known in the art, and may include apertures or pre-drilled holes to accept screws, bolts, disposable rivets, throw-pins or dowels, or other elongate fastening members, or tab and slot arrangements, press-fit couplings, cooperating threaded elements, or other connectors. It will be appreciated that lower rail assembly **670** may serve the dual purposes of adding horizontal bracing to column **320** (resulting in a more stable and rigid framework for supporting forms and reinforcing system components from rail assembly **500**) as well as securing a lower portion of recess form **601** (or any generic interior form **600**) to ensure proper vertical orientation of the structural member **200** to be constructed as substantially plumb, thereby ensuring that the bottom two corners of the structural member **200** to be fabricated are properly (and predictably) aligned.

[0090] In some instances when rail assembly **670** may be longer or shorter than a span between adjacent columns **320** (at corners of structural member **200**, for instance), such a generic lower rail assembly **670** may be used to join two or more adjacent exterior forms **800** (described below) and may be operative to prevent relative movement of these exterior forms **800** during use, either in addition to, or in lieu of a lower rail assembly **670** as implemented in FIGS. **6A** and **6E**. Though

this alternative is not illustrated in the drawing figures for the sake of brevity, it will be appreciated that such an arrangement may simultaneously accomplish two goals: first, it conveniently uses a standard, “universal” length of lower rail assembly **670**, rather than requiring “one-off,” specially designed assemblies to accommodate different required or desired lengths at a construction site; and second, it allows two adjacent exterior forms to be rigidly secured to each other, thus ensuring that each exterior form **800** is flush with the other, and prevents relative movement of each. In combination with tie assemblies that secure interior forms **600** with exterior forms **800** during use (described in detail below in connection with FIG. **8A**), this strategy may ensure that the structural member **200** to be fabricated is plumb and true when completed.

[0091] Upon installation of lower rail assembly **670** and suspension of recess form **601** from interior hanging chord **551**, a lower portion of recess form **601** may be attached, secured, or otherwise simply made to contact or abut support clips **673** disposed longitudinally along longitudinal axis **271** at various locations along lower rail **671**. The locations of support clips **673** may be selectively adjustable, for example, such that support clips **673** may be selectively relocated depending upon a structure of recess form **601** (or any other generic form **600**) that is to be supported. Additionally or alternatively, recess form **601** (or others described below) may be fabricated with cooperating structures to mate specifically with support clips **673** on a “universal” or “form agnostic” lower rail assembly **670**. In some instances, recess form **601** may be rigidly attached (such as by bolts, screws, press-fit couplings, etc.) to support clips **673**, though this is not strictly necessary. In some implementations, support clips **673** may be deployed at a point at which two adjacent forms (such as recess form **601**) abut each other (such as at bumper **640**), and may be configured and operative to prevent relative movement between the two abutting forms.

[0092] It is noted that any number of conventional mechanisms may be used to suspend recess form **601** (or any generic form **600**) from interior hanging chord **551**, and that the present disclosure is not intended to be limited by any particular mechanism, strategy, or arrangement of elements. By way of example, c-hooks, s-hooks, wires, cables, and the like may all be suitable; the type of hanging hardware, as well as its construction, materials, coatings, and the like, may be selected as a function of the construction of the forms **600** themselves (as well as their weight and the distribution of that weight), corrosion resistance, disposability or reusability, and a variety of other factors. In some cases, a simple bolt fastener may be used to suspend a form **600** from interior hanging chord **551**. FIGS. **6A** and **6D** through **6I** illustrate such hanging hardware or hanging hardware attachment point (reference numeral **641**), which may be as simple as a metal loop, hook, throw-pin, or dowel integrated into the structure of a form **600**, or it may be a pre-drilled hole for accepting a bolt, screw, throw-pin, dowel, or threaded eyelet, for instance, which may be employed to grasp a wire, cable, hook, or other appropriate cooperating hardware suspended from internal hanging chord **551** or another form **600** (such as at hanging hardware **642**) as set forth below.

[0093] In particular, the illustrated structural arrangements depict a reciprocating throw-pin **695** mechanism which may have utility in engaging recess form **601** (or any other form described herein) with interior hanging chord **551** (or from exterior hanging chord **552**, as the case may be). In this arrangement, throw-pin **695** is capable of linear translation along an interior of vertical member **652**, and may be made to project vertically through an aperture (reference numeral **643**) at the top of form frame **650**; a detent mechanism (such as a groove and flange arrangement, for example) may be employed to maintain throw-pin **695** in this extended position during use. As best seen in FIGS. **6A**, **6D**, **6E**, and **6H**, an horizontal flange at attachment hardware **641** may extend over internal hanging chord **551**, supporting the weight of recess form **601**; throw-pin **695** may then be raised to protrude out of aperture **643** to engage a downward projecting lip or other cooperating structure of interior hanging chord **551** to prevent unwanted movement of recess form **601** during construction operations. A handle **696** or other mechanism may be used to slide throw-pin **695** upward (to project from aperture **643**) or downward (to retreat into aperture **643**, i.e., back into vertical member **652** of form frame **650**) during installation and removal of recess form **601**

from interior hanging chord **551**. In some implementations, throw-pin **695** may be spring-biased, for example, causing it to retreat into aperture **643** or to extend therefrom (depending upon the nature of the bias) unless the bias is overcome deliberately by work crew manipulation; this may ensure proper positioning of throw-pin **695** during use and when a particular form is installed, removed, or stowed.

[0094] In addition to the trapezoidal shape of recess form **601**, a variety of other shapes and sizes for interior form **600** are contemplated. Corner forms (e.g., for use where two adjacent structural members meet at an acute, right, or obtuse angle), flat forms (e.g., for use at door and window locations, or in locations where no recess region **299** is desired), beam forms (e.g., for use at overhead beam locations at windows and doors), and sill forms (e.g., for use below window locations or at door thresholds) may all be used. As with recess form **601**, all of these additional form shapes and sizes may be pre-fabricated for efficient suspension from rail assembly **500** as set forth above or for support by another form **600** as set forth below. In some cases, simple bolt fasteners or wing nuts may be sufficient for hanging forms **600**, though other integrated hardware may be employed as desired.

[0095] In that regard, FIG. **6J** is a top cross-sectional view illustrating construction details of one implementation of a generic flat interior form **600**. It will be appreciated that all such forms may include a structural frame **650** having vertical members **652** (perhaps supporting linear translation of one or more throw-pins **695**) and cross members **630** to support skin **659**, in general, and beam webs **657**, in particular, to create a smooth interior surface **611** of the reinforced structural member **200** to be fabricated. Fabrication of structural form frame **650** and skin **659**, as well as implementation and operation of throw-pins **695**, may be substantially similar to the description of FIGS. **6H** and **6I** above with reference to recess form **601**.

[0096] Addressing general use of additional shapes of a form **600**, it is noted that FIG. **6D** is a side view of a flat beam form **602** suspended from interior hanging chord **551**; FIG. **6E** is a side view of flat beam form **602** similarly suspended from interior hanging chord **551**, and also shows a sill form **604** that may be supported by an abutting side of an adjacent form (not illustrated) and additionally selectively attached to lower rail assembly **670** at support clip **673** in some instances where desirable. It is noted that alternative approaches may be employed, such as by suspending a lower form **600** from a higher form, either in lieu of, or in addition to, supporting it from adjacent forms **600**. FIGS. **6F** and **6G** illustrate, respectively, simplified plan views of recess form **601** and flat beam form **602** (both sill form **604** and a full-height flat interior form **605** (see FIG. **6L**) may be similar to beam form **602** in plan view). An exterior surface **610** of these forms **601** and **602** is structurally significant for the forms **601** and **602** themselves (as this is part of the frame (such as form frame **650** in FIGS. **6H** and **6J**) that gives the forms strength and rigidity), but this exterior surface **610** has no bearing on the shape of the reinforced structural member **200** to be fabricated. Interior baseline surface **611**, on the other hand, defines an interior surface of the reinforced structural member **200** to be fabricated (having a baseline thickness defined, in part, by the location of baseline surface **611**); similarly, recess surface **621** (FIG. **6F**) creates recessed region **299** having recess depth **298** where no construction material can be deposited (in that regard, recess surface **621** may be analogous to that recess surface **297** discussed above in connection with FIGS. **1C** through **1E**).

[0097] In general, it will be appreciated that flat beam form **602** and sill form **604** may be selectively employed to support or otherwise to cooperate with dedicated inserts (described below) which define two different types of openings: a “peninsula” opening which creates a doorway (FIG. **6D**); and an “island” opening for a window (FIG. **6E**). As with recess form **601**, the various flat beam and sill forms **602** and **604** (as well as any other generic form **600**) may be fabricated with high-strength aluminum or steel elements that are structurally adjoined and braced to prevent deflection and twisting, such that the resulting rough opening size, angular orientation, and elevation defined by the inserts (described below) supported by the forms are precise from job site

to job site; such a frame structure may be used to support a light-weight fiberglass or composite skin as noted above in connection with recess form **601**, and as specifically described above with reference to form frame **650** and skin **659** illustrated in FIGS. **6H** through **6J**. Flat beam form **602** (see FIGS. **6D** and **6E**) may similarly be suspended from interior hanging chord **551** using attaching hardware **641** as set forth above. In the event that sill form **604** is to be placed below beam form **602** (see FIG. **6E**), such sill form **604** may be hung, braced, or otherwise rigidly supported by an adjacent form **600** (not illustrated in FIG. **6E**) on either side of sill form **604**. Additionally or alternatively, a lower sill form **602** may be supported by hanging hardware (such as **641**) from a point (reference numeral **642** in the drawing figures) at or near a lower portion of higher beam form **602**, though this is not necessary in many arrangements. Again, hanging hardware **641** may be embodied in or comprise fasteners such as wire loops, hooks, or other fasteners that may be integrated with the structural framework of a form **600**, or may be selectively inserted into pre-drilled, precisely located fastener holes. In some instances, it may be desirable to include turnbuckles, threaded members, ratcheted connectors, or chain links to enable precise placement of a top and bottom of a form **600** in the vertical direction. In the implementations illustrated in the drawing figures, an horizontal flange at attachment hardware **641** may extend over internal hanging chord **551**, supporting the weight of generic form **600**, and throw-pin **695** may then be raised to protrude out of aperture **643** to engage a downward projecting lip or other cooperating structure of interior hanging chord **551**; this same mechanism may be used to hang sill form **604** from a higher beam form **602**, for instance, though as noted above it may be preferable in most instances to attach sill form **604** rigidly or otherwise fixedly to an adjacent form **600** as set forth in more detail below.

[0098] Upon installation of any beam forms **602**, any sill forms **604**, or both, as desired or applicable, and after vertical placement adjustments, as desired or necessary (and as applicable), a finished rough opening defined by inserts supported by the forms may be created to be dimensionally precise in both principal directions (vertically and horizontally), angularly precise at every corner, and geometrically accurate in an elevation perspective. With the framework erected in accordance with the discussion above and the top rail assembly **500** in place to suspend suitably constructed forms **600**, each insert defining an opening for a given structural member or wall may be installed in a manner of minutes, regardless of opening size or location, without reliance upon surveyors, expensive (and perhaps fragile) equipment, and skilled labor.

[0099] It will be appreciated that forms **600** may be implemented to accommodate large window and door openings by placing dedicated panel forms at each side jamb, and a large opening beam at the top horizontal header. Application specific forms **600** may facilitate creation of large openings (sliding glass doors, large picture windows, garage doors, etc.), arched, circular, or irregularly shaped openings, and the like, with repeatable accuracy and mechanical efficiency substantially as set forth in more detail below.

[0100] In that regard, FIGS. **6D**, **6E**, **6K**, **6M**, and **6O** illustrate an arrangement to accommodate large spans for wide window or door openings, and FIGS. **6L** and **6N** illustrate an arrangement to accommodate shorter spans for regular or narrow fenestrations. For a wide opening, a beam insert (reference numeral **609**) may be employed to prevent construction material from entering the void created below. In particular, beam insert may be bolted, screwed, clipped, or otherwise fixed at attachment points **619** to a respective frame element (reference numeral **608**, not illustrated in FIG. **6K** for clarity); in use, frame elements **608** span at least a distance (labeled “h”) from a lower portion of beam insert **609** to an upper portion of sill form **604**, though they may extend all the way to foundation **295** in the case of a door opening (FIG. **6M**). Beam form **609** and its attendant frame elements **608** may prevent construction material from entering a void created below beam form **609** in the area intermediate beam form **602** and sill form **604**, or between beam form **602** and foundation **290**, in the case of a door where no sill form **604** is employed (see FIG. **6M**, which shows a simplified interior view of this arrangement for a wide door). For narrower window or

door openings, a full height flat form **605** may be hung from interior hanging chord **551**, and an appropriately sized opening insert **607** may be attached to flat form **605**, such as at attachment points **617**, and any adjacent or abutting inserts, such as at attachment points **619**. In this arrangement, opening insert **607** may prevent construction material from occupying the space defined by opening insert **607**, itself, thereby creating an opening in the reinforced structural member **200** to be formed (see FIG. **6N**, which shows a simplified interior view of this arrangement for a simple, narrow door, and FIG. **6L**, which shows a side view of installation of an opening insert (**607**) for use in connection with fabricating a simple, narrow window). Finally, it will be appreciated that beam insert **609** may be used to span adjacent opening inserts **607**, thereby making frame elements **608** unnecessary (see FIG. **6O**, which shows a simplified interior view of this arrangement for a wide window). As indicated in FIG. **6O**, use of sill frame **604** below beam insert **609** will allow a work crew member to reach into the cavity so created in order to ensure that construction material flows (in the direction of the arrows) to fill the entire span below even the widest of windows.

[0101] It will be appreciated that attachment of beam insert **609**, frame elements **608** (if applicable), and opening insert **607**, either to each other or to interior forms **600**, may be accomplished as noted above, i.e., with screws, bolts, clips, tie-rods, clamps, or other appropriately sized and dimensioned fastening members generally known in the art. It is further noted that the construction of beam insert **609**, frame element **608**, and opening insert **607** may be similar to the frame **650** and skin **659** of forms **600** described above, though simplified or conventional approaches may also be used. The present disclosure is not intended to be limited by any specific construction or framing techniques used for beam insert **609**, frame element **608**, or opening insert **607**, or by any particular manner in which the various system components are mechanically coupled (though it is noted that any hardware may be selected from materials having appropriate strength, ductility, and corrosion resistance characteristics that are necessary or desirable for a particular use, expected working environment, and longevity considerations).

[0102] Upon completion of the installation of the various system components, including appropriate interior forms **600** which define straight wall portions, recessed wall portions, and corners, as well as beam inserts **609**, opening inserts **607**, **608**, or both, as applicable, which define doors, windows, and the like (in accordance with a particular building's blueprint specifications), an entire interior face plane of that building's walls is defined.

[0103] As noted above with reference to other physical elements of a system for fabricating a reinforced structural member **200**, the specific materials (such as, for example, aluminum, steel, other metals or alloys, glass or ceramics, fiberglass or other composites, and the like) and dimensions of the forms illustrated in FIGS. **6A** through **6O** may vary as application-specific design choices, and the specifics discussed above were provided by way of example only. The present disclosure is not intended to be limited by the size, shape, profile or contour, or by the materials selected for implementation of, the forms **600** used to define the internal surfaces of the reinforced structural member to be fabricated.

[0104] FIGS. **7A** and **7B** show, respectively, a side view and a plan view of aspects of a reinforcing system for use in connection with a reinforced structural member. In some implementations, reinforcing system **700** may generally comprise three components: wall plane reinforcement element **791**, which may include vertical, horizontal, or hatched components; beam reinforcement element **792** (not shown in FIG. **7B** for clarity), and post reinforcement element **793**. These reinforcement elements **791** through **793** serve to provide internal resistance to axial, flexural, torsion, and shear stresses for reinforced structural member **200** as described above, and may be selectively suspended or otherwise supported from a reinforcement hanger **771** through **773**, as applicable, in appropriate locations as set forth below.

[0105] Specifically, in one implementation, a reinforcement assembly **770** may generally comprise an outer hanger **771**, a center hanger **772**, and an inner hanger **773**. As best illustrated in FIG. **7A**,

hangers 771 through 773 may be selectively attached to buttons, slots, hooks, or other hardware integrated with or otherwise attached at specific locations on cross member 550 element of rail assembly 500. It is noted that the illustration depicts three columns of such hardware integrated with cross member 550 (see reference numerals 581 through 583 in FIGS. 5A, 6A, 6D, and 6E), with outer hanger 771 occupying the right column 581 of attachment points, center hanger 772 occupying the center column of attachment points 582, and inner hanger 773 occupying the left column of attachment points 583. In some instances, inner column 583 may remain unused (i.e., no inner hanger 773 need be deployed) as illustrated in FIG. 7A, but the symmetrical structure allows rail assembly 500 to be installed upon and secured to top support structure 400 in either direction, i.e., it is not possible to engage or otherwise to install rail assembly 500 onto top support structure 400 in a “backwards” configuration or orientation. In some other instances, however, such as when a beam portion of a reinforced structural member 200 requires two substantially parallel bars or other reinforcing elements (e.g., running along longitudinal axis 271), it may be necessary or desirable to use inner hanger 773 at the left attachment point, i.e., inner column 583, for one such beam reinforcement element 792, and center hanger 772 or outer hanger 771 for the other such beam reinforcement element 792 (e.g., to satisfy building codes, local ordinances, construction industry “best practices” guidelines, and the like). Otherwise, it may be desirable that a single beam reinforcement element 792 is supported by center hanger 772 as depicted in FIG. 7A.

[0106] The location of these attachments points (columns 581 through 583) on cross member 550 ensures that center hanger 772, once coupled with center column 582 of attachment points, will always hang directly over a center of the baseline thickness of structural member 200 (i.e., a lateral distance $T/2$ from interior baseline surface 611); thus, a single beam reinforcement element 792 supported by center hanger 772 will always be at the lateral center of a beam structure having the full baseline thickness T . Depending upon the viewpoint of top rail assembly 500, either the left column 583 of attachment points (the left in FIG. 7A) or the right column 581 of attachment points (the right in FIG. 7A) may be used to ensure that outer hanger 771 will hang between recess surface 621 (see also reference numeral 297 in FIGS. 1C through 1E) and an exterior surface (see reference numeral 130 in 1C through 1E) of the reinforced structural member 200 to be fabricated as depicted in the drawing figures.

[0107] Hangers 771 through 773 may be attached to cross member 550 via any conventional attaching hardware generally known in the art. Press-fit couplings, set screws, bolts, rivets, and the like are all contemplated. In some instances, attachment points on cross member 550 may be pre-drilled holes to accommodate screws or bolts, for example, or corresponding protuberances extending from hangers 771 through 773. Alternatively, hangers 771 through 773 may incorporate pre-drilled holes to engaged nubs or other protrusions of respective columns 581 through 583 at attachment points on cross member 550. In that regard, hangers 771 through 773 may be fabricated using disposable or recyclable plastic, rubber, silicon, or other material, though metals, alloys, fibers, and composites are also contemplated. The present disclosure is not intended to be limited by the nature of the materials used for fabricating hangers 771 through 773, or the manner in which these components are attached to, coupled with, or suspended from attachment points on cross member 550, though it is noted that the foregoing lateral placement of outer hanger 771 and inner hanger 773 relative to center hanger 772 may be of importance in many applications.

[0108] Specifically, selective placement of outer hanger 771 and inner hanger 773 along transverse axis 272 will influence the respective lateral positions of wall plane reinforcement element 791 and any beam reinforcement element 792 relative to the overall structure of reinforced structural member 200; this is particularly true when two beam reinforcement elements 792 are supported, respectively, by inner hanger 773 and outer hanger 771, rather than in the case of a single beam reinforcement element being supported by center hanger 772.

[0109] By way of additional background, it is noted that a wall's (or other structural member's) reinforcing system may be one of its most structurally important components (depending upon the

application), and is typically one of the most expensive components, taking into consideration both material costs as well as labor costs associated with its installation. Accordingly, deliberate control over the amount of materials (such as steel) employed in a reinforcing system (such as reinforcing system **700**), and carefully minimizing labor time during installation of same may benefit overall construction economies. Hanging reinforcement system **700** from top rail assembly **500** as indicated in FIG. 7A may generally be completed after installation of all interior forms **600** so that the overall geometry and openings of structural member **200** are already precisely defined to allow for a properly reinforced assembly. There are basically three types of reinforcement needed to reinforce structural member **200** in accordance with ACI requirements and structural norms: wall plane reinforcement (see, e.g., wall plane reinforcement element **791**, which may include vertical, horizontal, or hatched components); beam reinforcement over openings (see, e.g., reference numeral **792** in FIG. 7A); and column or post reinforcement (see, e.g., post reinforcement element **793**).

Wall Plane Reinforcement:

[0110] In typical concrete wall construction, horizontal and vertical steel bars (i.e., rebar) are placed and oriented, individually, and thereafter secured to each other using wire ties to create a steel lattice. Since each bar must be placed individually, and subsequently attached manually to a number of other bars that cross it, the process of creating this latticework arrangement is both labor intensive and prone to error and irregularities. This increases construction costs, because skilled labor is involved—and the process itself demands a lot of that skilled labor.

[0111] In a departure from conventional methodologies, on the other hand, wall plane reinforcement element **791** may eliminate rebar and wire ties altogether, replacing them with one or more pre-fabricated components. In some implementations, wall plane reinforcement element **791** may be embodied in or comprise a pre-fabricated welded wire fabric sheet, for example, though other options are contemplated. For example, wall plane reinforcement element **791** may be embodied in or comprise a pre-fabricated fiberglass or composite material mesh, sheet, or latticework; specifically, wall plane reinforcement element **791** need not be fabricated of steel or other metals, though metal reinforcement is popular in the industry and specified in some building codes and industry standards or guidelines. The specific construction, gauge, thickness, orientation, and pattern of metal or fabric sheet may vary by application or as a design choice, but in any event, wall plane reinforcement element **791** may be easier to manipulate and to install than conventional rebar and wire tie implementations.

[0112] For example, ACI requires that a wall must be reinforced with a certain percentage of steel (or other suitable metal or composite material) per area of concrete; in accordance with one such standard, the specified percentages are 1.2% vertical steel to concrete area and 2.0% horizontal steel to concrete area. In accordance with aspects of the systems and methods set forth herein, however, since the unique cross-sectional shape of structural member **200** includes recessed regions **299** having only about one half the concrete thickness of the baseline thickness, this steel area requirement specified by the applicable standard is reduced by about half. As a result, instead of bulky rebar that is cumbersome to handle and to install, pre-fabricated lightweight reinforcing steel mesh sheets (or other suitable reinforcing webbed or meshed materials) may be utilized to satisfy the applicable reinforcing requirements specified in the standard. Further, due to the fabrication system framework's top rail assembly **500**, various components of reinforcing system **700** may be installed by hanging the reinforcing steel (such as wall plane reinforcement element **791**) from cross members **550** with the use of inexpensive plastic support hangers (such as outer hanger **771**). As noted above, hangers **771** through **773** may have connection stubs or other protuberances that are simply snap-fit or otherwise inserted quickly and efficiently into pre-drilled, precisely located holes in cross members **550**, thereby ensuring that the various reinforcing elements **791** through **793** may be precisely located within an inner core of structural member **200**. While other embodiments are contemplated for such hangers **771** through **773** (and the mechanical connections

between them and the attachment points associated with cross member **550**), it may be desirable to use inexpensive, disposable, or consumable hanger hardware for this limited purpose. Further, since the reinforcing sheet or lattice (e.g., wall plane reinforcement element **791**) is suspended from above rather than supported at foundation **290**, all proper ACI concrete coverage requirements are satisfied around the entire perimeter of reinforcement system **700**, even at the bottom.

Beam Reinforcement:

[0113] In accordance with many applicable building codes or relevant industry standards, beams (i.e., portions of a structure above a fenestration such as a door or window) may be reinforced with the use of longitudinal bars and stirrups (if necessary for shear forces). However, given the subject matter disclosed herein, it will be appreciated that wall plane reinforcing concepts in reinforcing system **700** may be utilized for both flexural and shear support, thereby reducing the conventional beam bar reinforcing required to resist loads. In this case, beam steel may be installed by hanging longitudinal bars or other suitably modified wall plane reinforcement element **791** material from top rail assembly **500** (e.g., adjacent beam forms **602**); this component is illustrated in FIG. 7A at reference numeral **792**, and it is reiterated that such beam reinforcement element **792** may be embodied in or comprise a mesh material (e.g., similar to that of wall plane reinforcement element **791**), or it be embodied in or comprise rebar or other steel, alloy, or composite rods, cylinders or other suitably sized and dimensioned elongate members configured and arranged to run horizontally along longitudinal axis **271**. Vertical support for beam reinforcement element **792** may be accomplished with the use of any of the various implementations of hangers **771** through **773** described above. Again, as noted above, during use, any such suitable hanging hardware may be so positioned on cross member **550** to ensure that the beam steel (or other material) of beam reinforcement element **792** is precisely located within a core of the structure/beam, thus enabling precise, repeatable, and efficient installation. In some instances, for example, such a beam reinforcement element **792** may be hung from center hanger **772** as depicted in FIG. 7A, ensuring that the reinforcement is at a centerline (T/2) of the baseline thickness of the reinforced structural member **200** to be fabricated. Alternatively, in circumstances where more than one longitudinal bar or other beam support material is required or desired for beam reinforcement, then these multiple beam reinforcement elements **792** may be symmetrically hung from inner hanger **793** and outer hanger **771**, and possibly from center hanger **772** as well, as desired.

Column Reinforcement:

[0114] Columns may generally be reinforced with the use of vertical bars, posts, tubes, or other elongate structural members; such a “post reinforcement element” is illustrated abstractly at reference numeral **793** in FIG. 7A, and its location is indicated in the plan view of FIGS. 6B and 7B. In this context, it will be appreciated that the term “column” is intended to be broad enough to include horizontal runs of a wall or other structural member **200** that are at the baseline thickness (i.e., areas that do not include a recessed region **299**). Due to the novel corrugated, crenelated, or rippled geometry of structural member **200** overall, the column portions that do not include a recessed region **299** may be reinforced with less steel or other reinforcing elements than in conventional designs (such as wall **100** in FIGS. 1A and 1B), and the minimal post reinforcement element **793** may be positioned in a more strategic location than in traditional concrete/masonry construction, while still providing superior load resistance performance for the overall structural member **200**. Post reinforcement element **793**, positioned as indicated in the drawing figures, may provide support in resistance of vertical (i.e., column and axial) loads, and may also assist with resistance of horizontal loads that cause tension on an interior face of the reinforced structural member **200** to be fabricated as discussed above in connection with FIGS. 1C through 1E.

[0115] In some implementations, post reinforcement element **793** may be embedded into foundation **290**, or otherwise rigidly affixed to it, as in conventional wall design and construction. Additionally or alternatively, post reinforcement element **793** may be suspended from the fabrication system framework's top rail assembly **500**, for example, either from inner hanger **773** or

center hanger **772**, depending upon application. In instances where post reinforcement element **793** is integral with or otherwise rigidly attached or coupled to foundation **290** from below with conventional hardware, it may be desirable not to support it from above, such that the entire post reinforcement element **793** will be covered with construction material upon completion of the reinforced structural member **200**.

[0116] Post reinforcement element **793** may be embodied in or comprise a solid cylinder, post, or bar, a hollow tube, or other elongate structural member, and may have an exterior surface that is smooth, roughed, grooved, or fluted, for example, to facilitate mechanical bonding with or adhesion to concrete or other materials used for the structure of structural member **200**. It will be appreciated that post reinforcement element **793** may have any plan cross-sectional shape that is appropriate for a given application, and it may be installed by simply tying it to pre-installed dowels located at foundation **290**, for instance, and to the wall/beam top bar or it may be hung from a hanger **773** or **772** as indicated above. Inexpensive, plastic spacers (of any selected geometry) may be installed at selected vertical locations of post reinforcement element **793** to ensure precise placement, and to ensure compliance with all ACI and applicable building code concrete coverage requirements.

[0117] In accordance with the foregoing mechanical framework, optimized structural design, and construction methodologies focused on efficiency, structural member **200** may be laid out and wall plane reinforcement elements **791**, beam reinforcement elements **792**, and post reinforcement elements **793** may be placed and installed to ensure code-compliant reinforcement in a precise, repeatable, and efficient manner, all while employing much less steel (or other reinforcing material) and much less labor than required by traditional construction methods fabricating conventional, inefficient structural members (reference numeral **100**). Using the systems and methods disclosed herein, an entire span of a large horizontal wall run may be reinforced within a matter of minutes, using approximately one-third less reinforcing material than is normally required for conventional applications, while still producing a structurally superior installation.

[0118] FIG. **8A** shows a side view of one implementation of a form selectively suspended from the rail assembly of FIGS. **5A** through **5C** and operative to define an exterior surface of a reinforced structural member. An exterior form **800** may be embodied in or comprise a flat form **801** having a baseline surface **811** that, along with baseline surface **611** of interior form **600**, defines a baseline thickness of structural member **200**. Specifically, since flat form **801** is an exterior form, it is illustrated as being suspended from exterior hanging chord **552** coupled to cross member **550** of rail assembly **500**.

[0119] In some instances, conventional construction forms may be used for exterior form **800**. Alternatively, an exterior form **800** such as flat form **801** may be constructed in the same manner, and of the same materials, as interior forms **600** described above in connection with FIGS. **6A** through **6O**, and in particular, the detailed discussion of the form frame **650** and skin **659** in connection with FIGS. **6H** through **6J**. In one implementation, for example, exterior form **801** may be embodied in or comprise a frame (such as form frame **650**) fabricated from high strength structural aluminum (or other metal or composite materials) with framing components precisely placed to maximize strength, minimize deflections, and reduce weight; as set forth above, such a form frame **650** may generally support a surface or “skin,” (reference numeral **659**) which may be fabricated from ultra-high strength, lightweight fiberglass (or other composite material) having a “T” beam cross-sectional profile (e.g., beam webs **657** made to engage cross beams **630** of frame **650**). In this implementation, as illustrated in FIG. **8A**, cross beams **830** correspond to cross beams **630** described above, and skin **659** (e.g., defining baseline surface **811**) is applied or otherwise attached to the frame cross members such that each frame component (including vertical members such as **652**, not shown in FIG. **8A**) and each cross beam **830** is precisely placed to be structurally loaded to capacity during use to create even and uniform and insignificant skin deflection, even though form **801** must withstand a triangular load distribution (up to 1,400 pounds per square foot,

for a 10 foot high wall, for example) when wet construction slurry is poured.

[0120] As noted above, the aforementioned skin **659** of flat form **801** generally comprises a baseline surface **811** to define an exterior surface, and thereby (and in cooperation with baseline surface **611** of an interior form **600**) the baseline thickness **T**, of reinforced structural member **200**. Flat form **801** may be so dimensioned to span an horizontal run substantially the same as that of an opposed interior form **600**. As best illustrated in FIG. **8A**, with flat form **801** suspended from exterior hanging chord **552** opposed to recessed form **601** suspended from interior hanging chord **551**, wall plane reinforcement element **791** (when suspended from outer hanger **771**) is disposed intermediate baseline surface **811** and recess surface **621**—thus positioning wall plane reinforcement element **791** in the structural “flange” of structural member **200**, where it may be entirely encased in concrete or other construction material.

[0121] Similar to FIG. **6A**, FIG. **8A** also illustrates a plurality of cross beams **830** integrated with or otherwise supporting the skin **659** (and in particular, baseline surface **811**) of flat form **801**. Cross beams **830** are generally elongate structural elements, running horizontally along longitudinal axis **271**, to provide support for skin **659**, and in particular, interior surface **811**, of form **801**, and may therefore be distributed vertically within the frame structure (see FIG. **6H**) as needed to accommodate a varying force profile. Accordingly, vertical distribution of cross beams **830** may be more dense at a lower portion of flat form **801** than at upper portions above. The specific number and vertical distribution of cross beams **830** may be application-specific as noted above, and may vary as desired to satisfy particular requirements or design preferences; as was true in FIG. **6A**, the arrangement depicted in FIG. **8A** is provided by way of example only. As with interior forms **600**, exterior forms **800** may be made to abut each other using bumpers or other mechanisms (see reference numeral **640** in FIGS. **6A** and **6I**) that will accommodate slots (reference numeral **890**) to facilitate coupling with interior forms **600** as set forth below. As noted above with reference to interior forms **600**, such bumpers for exterior forms **800** may also be fabricated of rubber, silicone, plastic, or other pliable material, and may include slots or other apertures to accept a sleeve **881** and bolt **891** for a tie assembly **880** that secures exterior form **800** to interior form **600**, and may also serve to secure adjacent forms (either interior forms **600** or exterior forms **800**) together.

[0122] In addition to the full length flat form **801** (i.e., “full length” in this context is intended to mean that it spans an entire vertical height of structural member **200**), a variety of other shapes and sizes for exterior form **800** are contemplated. Corner forms (e.g., for use where two adjacent structural members meet at an acute, right, or obtuse angle), beam forms (e.g., for use at overhead beam locations at windows and doors), and sill forms (e.g., for use below window locations and at door thresholds, if applicable) may all be used. As noted above in connection with interior forms **600**, such complementary shapes and sizes of exterior form **800** may be pre-fabricated for efficient suspension from rail assembly **500**, or from another form **800**, as set forth above. In some cases, simple bolt fasteners or wing nuts may be sufficient for hanging forms **800**, though other integrated hardware may be employed as desired and as set forth in detail above. Specifically, attachment hardware **841** and throw-pin **895** may be analogous to attachment hardware **641** and throw-pin **695** described above, and may be operative in a substantially similar manner, to support exterior forms **800** from hanging chord **552**.

[0123] Addressing additional shapes of a form **800**, it is noted that the forms **602** and **604** illustrated in FIGS. **6D**, **6E**, and **6G** may all have counterpart exterior forms **800**, as is generally known in the art, such that an exterior baseline surface **811** may define an exterior surface of the reinforced structural member **200** to be fabricated opposite a counterpart interior form **600**. Additional details have been omitted for brevity, but will be readily apparent from the disclosed subject matter. As an alternative, a system may implement exterior forms **800** that are each embodied in a “full length form” (that spans an entire vertical height of structural member **200**) at every point along longitudinal axis **271**, for simplicity; in this case, all openings or fenestrations are defined exclusively by inserts **607** or **609** (and possibly framing components **608**), as the case may

be, without requiring different types of exterior forms **800**.

[0124] In addition to cross beams **830** that provide rigidity, flat form **801** may also include through-bores or slots (reference numeral **890**) at locations opposite cooperating through-bores or slots in recess form **601** (reference numeral **690** in FIG. 6A) or any generic interior form **600**. As noted above, these may be positioned at a point where adjacent forms **800** abut each other, such as at a rubber, silicone, or other flexible bumpers, and may generally be aligned to cooperate with those slots **690** on interior forms **600**. In use, slots **690** and **890** allow for insertion of tension ties or similar devices which prevent relative movement between recess form **601** and opposing exterior form **801** during use.

[0125] Specifically, an external form **800** may be secured to an interior form **600** with the use of a tension tie assembly (reference numeral **880** in FIG. 8A). In some implementations, tension tie assembly **880** may be so designed and constructed as to be reusable from structure to structure (or from job site to job site), which represents a significant departure from conventional concrete wall or other form systems. This may be accomplished with the use of inserts that span a distance from interior surface **611** to exterior baseline surface **811** using an insert, or sleeve **881**, embodied in or comprising a tube of sufficient internal diameter to accommodate a bolt **891** that extends from exterior form **800** to interior form **600**. The inserts (such as sleeve **881**) may be inexpensive, and constructed of plastic, polyvinyl chloride (PVC), acrylic, silicone, or other suitable material, depending upon corrosion resistance properties, cost, and specific application requirements. This strategy accomplishes two important goals: first, it effectively creates a tunnel or conduit through the core of structural member **200** and through which bolt **891** (or a screw, cable, tether, or other elongate fastening member associated with tension tie assembly **880**) may be inserted; and it provides an inner spacer element that forces structural member **200** to be constructed at exactly the correct (and uniform) design thickness by establishing a connection between interior form **600** and exterior form **800**. Further, tension tie assembly **880** may be installed with these pre-fabricated, interchangeable components very quickly on a job site. In some implementations, tension tie assembly **880** includes a long bolt **891** (one bolt head is shown at reference numeral **890** in FIG. 8A) with suitably sized washers to secure interior form **600** and exterior form **800** via sleeve **881**, though other tension members such as cables, turnbuckles, and elastic elements may also be selected as a function of application specific parameters.

[0126] In some implementations, a tension tie assembly **880** may generally comprise a sleeve **881** having a flange or a radial widening at each end. Such a flange or flute may prevent concrete or other construction material from entering the tube created by sleeve **881** (and from escaping through slot **690**, **890**) during use, and, when sleeve **881** is of a specific length (for instance, “T”), then such a flange or flair will also help to ensure that the reinforced structural member **200** is always the correct thickness (i.e., a flanged sleeve **881** may also act as a spacer) by suitably positioning exterior form **800** at a known distance from interior form **600**. Additionally, it may be desirable that tension tie assembly **880** includes a C-shaped clamp at each end (see reference numeral **888** in FIG. 8A); in such an embodiment, one side of C-shaped clamp **888** may be made to engage a vertical member (such as **652**) of the form frame **650** of one form, while the other side of the clamp **888** may be made to engage a vertical member (such as **652**) of the form frame **650** of the adjacent form. Accordingly, tension tie assembly **880** with such a clamp **888** may not only connect form **600** and form **800** at an appropriate distance, but may also serve to ensure that adjacent interior forms **600** or adjacent exterior forms **800** do not move (either laterally or longitudinally) with respect to one another. In some implementations, clamp **888** may also serve as a washer, preventing a head of bolt **891** from slipping into slot **890**, which may create spacing errors between interior form **600** and exterior form **800**.

[0127] As noted above with reference to other physical elements of a system for fabricating a reinforced structural member, the specific materials (such as, for example, aluminum, steel, other metals or alloys, glass or ceramics, fiberglass or other composites, and the like) and dimensions of

the forms **800** described above in connection with FIG. **8A** may vary as application-specific design choices, and the specifics discussed above were provided by way of example only. The present disclosure is not intended to be limited by the size, shape, profile or contour, or materials selected for implementation of the forms **800** used to define the exterior surfaces of the reinforced structural member to be fabricated.

[0128] FIG. **8B** shows a side view of one implementation of a trolley supported by the rail assembly of FIGS. **5A** through **5C** and operative to deposit a slurry of construction material to form a reinforced structural member. Trolley **860** may generally include a hopper **869** that is configured and operative to contain concrete or other material slurry for use in fabrication of reinforced structural member **200**. Hopper **869** may include a sloped surface **863** that is designed to feed slurry to a funnel **865** which exits hopper **869**. Trolley **860** may be designed to have a track such that wheels **861** are spaced to engage top chords **521** and **522** of top rail assembly **500**.

[0129] During use, concrete or other construction slurry may be supplied to hopper **869** of trolley **860** with a traditional “pump truck” generally known in the construction arts. However, trolley **860**, capable of linear translation along top chords **521** and **522** of top rail assembly **500**, represents an innovation that facilitates efficient and proper placement of the slurry within the wall system assembly (i.e., between interior forms **600** and exterior forms **800**) without spillage. As illustrated in FIG. **8B**, trolley **860** may be so dimensioned that wheels **861** ride on a track created by top chords **521** and **522**, allowing linear translation for deposition of slurry all along longitudinal axis **271** of structural member **200**, without requiring repositioning of the pump truck or the pump truck hose. As concrete is dispensed from funnel **865** to the cavity between interior forms **600** and exterior forms **800**, trolley **860** may be made to traverse top chords **521** and **522** in a controlled manner, enabling construction slurry (such as concrete, for example) to be properly placed with little waste and minor errant mess.

[0130] It will be appreciated that control of trolley **860** may be manual in some instances, for example; additionally or alternatively, trolley **860** may be controlled in part (or entirely) by remote control. Simple wired or wireless (e.g., radio frequency (RF), near-field communications (NFC), or other wireless network) systems may be used to control a suitably configured trolley **860**. A battery or other power supply, a radio transceiver, and simple electronics (not shown in the drawing figures) may be incorporated into trolley **860** without undue experimentation or effort, as is generally known in the art of small radio-controlled toys, drones, or other apparatus.

[0131] FIG. **9** is a simplified functional flow diagram illustrating aspects of one implementation of a method of fabricating a reinforced structural member. As noted above, execution of a method such as that illustrated in FIG. **9** may not only result structural members that satisfy specific design and load resistance criteria, but also may reduce resources and time required by construction crews in the field to construct such structural members.

[0132] In accordance with the FIG. **9** implementation, a method **900** may generally begin with securing a base plate to a foundation or floor support structure as indicated at block **901**. As noted above, such an operation may involve securing an elongate base plate (such as base plate **220**) to a foundation which is operative to support the structural member (such as structural member **200**). In some implementations, such securing may generally comprise selectively attaching the elongate base plate to the foundation in an orientation that is substantially perpendicular to a longitudinal axis of the structural member (e.g., longitudinal axis **271** in FIG. **1C**). It is noted that the term “foundation” in this context is intended to be broad enough to include a floor structure, such as a joist, sill plate, beam, or joist hanger, for instance (see, e.g., the discussion of foundation or floor support element **290** above in connection with the discussion of FIG. **2A**). The present disclosure is not intended to be limited by the nature, structural characteristics, or implementation details of the foundation, and it is noted that any suitable support structure, including those elements comprising parts of a lower floor of a multi-story building, are contemplated by the operation depicted at block **901**.

[0133] Method **900** may proceed by attaching a column to a portion of the elongate base plate and bracing it in a substantially vertical orientation as indicated at block **902**. It will be appreciated in this context that the column may have a height selected to accommodate a height of the structural member to be fabricated. As noted above in connection with FIGS. 2A through 3B, some such operations may generally include attaching a column (e.g., reference numeral **320**) to a specific structure of the elongate base plate (e.g., column support bracket **221**) that is proximate to the structural member and configured and operative to engage a cooperating structure near a bottom portion of the column; bracing the column in a substantially vertical orientation may generally comprise using a brace member (e.g., reference numeral **340**) that is attached to a specific portion of the elongate base plate (e.g., brace member support flange **222**) that is distal to the structural member, and additionally attached to the column at a location above the foundation and below the height of the structural member (such as at a brace member attachment element **350** described above with reference to FIG. 3A). Additionally or alternatively, bracing the column may include attaching such a brace member to a brace support element that is independent of the elongate base plate, such as a brace shoe (reference numeral **240**) described above and best illustrated in FIG. 3B; in this case, the brace member may still be attached to the column above the foundation and below the height of the structural member using suitably cooperating mechanical couplings, such as brace member attachment element **350**. In the case where both an elongate base plate **220** and an independent brace shoe **240** are employed, the column may be braced in two substantially perpendicular directions as illustrated in FIG. 3B, facilitating orientation of the column in a substantially vertical (i.e., plumb) orientation.

[0134] Method **900** may continue by attaching a top support to the top of the column as indicated at block **903**. It will be appreciated that the “top” of the column in this context may generally be considered to be the free end opposite the elongate base plate or the foundation. As noted above, in some instances, it may be desirable to secure the top support (such as support **400** in FIG. 4A) such that a horizontal support flange (such as a cantilevered support **421** or **422**) is positioned in an orientation substantially parallel to a direction of the elongate base plate, or orientated along a transverse axis (such as indicated at reference numeral **272**) that is perpendicular to the longitudinal axis (reference numeral **271**) of the structural member to be fabricated.

[0135] In this case, a rail assembly (such as depicted at reference numeral **500**) may be selectively attached to the top support in a substantially horizontal orientation and substantially parallel to the longitudinal axis **271** of the structural member to be formed, as indicated at block **904**. As described in detail above, this rail assembly may be configured and operative to suspend the form structures that may be used to define the structural member to be fabricated; it may also support a trolley, hopper, container, or other movable dispensing component that may translate linearly along the rail assembly to deposit construction materials for the structural member as set forth in detail above. Where structural members meet at a corner of a structure, one rail assembly for a first structural member may be cantilevered to extend beyond the other rail assembly for a second structural member; because the form elements are selectively suspended from the rail assembly (rather than built up from the foundation), it is not necessary that every rail assembly be at an identical height above the foundation to ensure a properly constructed structural member or to ensure consistency from one structural member to another in the same building, yet it nevertheless may be desirable that intersecting rail assemblies are made to abut each other at the same height, rather than overlaying one above the other.

[0136] As indicated at block **905**, the method may continue by selectively suspending an interior form element (such as form elements **601**, **602**, and **604** in FIGS. 6A through 6E, for example) from the rail assembly to define an interior surface of the structural member, selectively suspending an exterior form element (such as form element **801** in FIGS. 8A and 8B, for example) from the rail assembly to define an exterior surface of the structural member, and (optionally) selectively suspending an opening form element from the rail assembly or from an adjacent form to define an

opening in which no structural material will be present in the structural member to be fabricated.

As noted above, it is not necessary to suspend an opening form element from the rail assembly itself; in many circumstances, it may be desirable to attach or suspend such an opening form element (see, for example, the “inserts” at reference numerals **607** and **609**) from an interior or exterior form itself, either in addition to or in lieu of supporting it directly from a rail assembly. [0137] As set forth above, the exterior form element and the interior form element may be selectively suspended from the rail assembly a distance apart, the distance between the interior form element and the exterior form element defining a baseline thickness of the structural member. Where a recessed region is included in the structural member, the interior form element may comprise a recess element (see, e.g., interior form **601** and recess element **620** in FIGS. **6A** and **6F**), having a recess depth (reference numeral **298** in FIG. **1D**) from the interior surface of the structural member, that creates the recessed region having an effective thickness ($T_{sub.e}$ in FIG. **1D**) that is less than the baseline thickness of the structural member.

[0138] As indicated by the dashed loop (reference numeral **995**), the operation (or selected portions of the operation) depicted at block **905** may be iterated any number of times as desired or necessary to set the forms for fabrication of a structural member (such as reinforced structural member **200**) having desired or required dimensions, specifically a length, such as along longitudinal axis **271**.

[0139] In some implementations, method **900** may continue by selectively suspending a reinforcing element (such as wall plane reinforcement element **791**, beam reinforcement element **792**, post reinforcement element **793**, or some combination of these) from the rail assembly intermediate the interior form element and the exterior form element as indicated at block **906**. It may be desirable that this operation comprises positioning the applicable reinforcing element in a position relative to the exterior form element to support the structural member against a bending moment. As noted above, biasing a reinforcing system **700** (such as may comprise wall plane reinforcement element **791**) towards the exterior surface may allow removal of material, enabling a structural member to take advantage of a recessed region and economizing on materials while still providing adequate structural support for both vertical (axial) loads as well as horizontal (lateral) loads. This arrangement is best illustrated in FIGS. **1D** and **8A**.

[0140] As indicated by the dashed loop (reference numeral **996**), the operation (or selected portions of the operation) depicted at block **906** may be iterated any number of times as desired or necessary to arrange sufficient reinforcing elements **791**, **792**, and **793** of reinforcing system **700** for fabrication of a structural member (such as reinforced structural member **200**) having desired or required dimensions, specifically a length, such as along longitudinal axis **271**.

[0141] Method **900** may conclude by depositing a slurry of construction material in a cavity between the interior form element and the exterior form element to form the structural member having the selected height as indicated at block **907**. As noted above, this deposition may include use of a trolley configured and operative to ride along portions of the rail assembly. Linear motion of the trolley along the longitudinal axis of the structural member to be fabricated, as well as deposition rate of slurry from the trolley, may be controlled manually or electronically, for example, in situations in which the trolley is equipped with appropriate control electronics, motors, and a power source. In some instances, it will be desirable to keep the structural framework created by method **900** in place for the duration required for the construction material to set or cure. As noted above, the construction material slurry may be concrete, and curing duration may be a function of the recipe used, atmospheric conditions such as temperature and relative humidity (which may be affected by altitude, for instance), the manner in which the slurry was mixed and the age of the slurry before it was deposited at block **907**, or a combination of these and a variety of other factors. The structural framework constructed in accordance with method **900** may be broken down when the construction material is sufficiently cured (not illustrated in FIG. **9**).

[0142] It is noted that the arrangement of the blocks and the order of operations depicted in FIG. **9** are not intended to exclude other alternatives or options. For example, the operations depicted at

blocks **901** through **904** may be reversed (or otherwise modified) in order, or they may be made to occur substantially simultaneously in some implementations. In some situations, it may be desirable to intertwine the operation depicted at block **906** with some of the operations depicted at block **905**, for instance. For example, it may be desirable, though not necessary, to suspend the reinforcing elements (at block **906**) prior to or substantially concomitantly with suspending the exterior form elements (at block **905**). Those of skill in the art will also appreciate that the operations depicted at blocks **906** and **907** may be occurring substantially simultaneously as well, and that the foregoing subject matter is susceptible of various design choices that may influence the order or arrangement of the operations depicted in FIG. **9**.

[0143] The foregoing system and method selectively place construction resources where they may be most structurally efficient. Specifically, a fabricated wall having selectively shaped recessed regions of a desired geometry may eliminate, or at least minimize, use of unnecessary concrete; these recessed regions also require placement of reinforcing elements (i.e., steel, fibers, fabric, or a combination thereof, for instance, such as illustrated in FIGS. **1D** and **8A**) in a more strategic, efficient location than at a structural member's center. Instead, a centerline of reinforcing elements may be positioned at or near a wall surface, so that the “d” value is maximized, thereby maximizing the moment resistance capability of the wall, overall. Moreover, the ACI-318 code specifies that, to ensure compliance, an amount of reinforcing steel (or other material) that is installed in a wall must be a fixed percentage of the wall's gross concrete area. According to the ACI-318 code, therefore, eliminating unnecessary concrete may also result in a corresponding percentage reduction of reinforcing material that must be used. As set forth above, such reduction in construction materials may be accomplished without compromising structural integrity of a wall so constructed, and while actually increasing its resistance to bending moments. In some implementations, either at a construction site or during the design phase, appropriate dimensional characteristics of a wall may be optimized to minimize the unit quantity of the two main materials used in the field of reinforced concrete or reinforced masonry wall construction: concrete or masonry; and reinforcing steel or other supporting material. As a result, over 30% of the concrete and 30% of the reinforcing material may be eliminated versus conventional wall construction, yet superior structural capabilities may nevertheless be achieved.

[0144] Several features and aspects of a structural member, as well as systems and methods of fabricating same, have been illustrated and described in detail with reference to particular embodiments, arrangements, or implementations by way of example only, and not by way of limitation. Those of skill in the art will appreciate that alternative implementations and various modifications to the disclosed subject matter are within the scope and contemplation of the present disclosure. Therefore, it is intended that the present disclosure be considered as limited only by the scope of the appended claims.

Claims

1. A structural member comprising: a first surface oriented to be substantially vertical, having a first height, and positioned to be exposed to an exterior side of a building; a second surface oriented to be substantially vertical, having a second height equal to the first height, and positioned substantially parallel to and in a spaced-apart relationship with the first surface, the spaced-apart relationship defining a substantially uniform baseline thickness of the structural member; a recessed region formed in a portion of the second surface and extending toward the first surface, the recessed region having a recess height that is less than the second height, a recess width that spans a horizontal distance of the second surface, and a recess depth from the second surface that is less than the baseline thickness of the structural member; wherein the recessed region defines an effective thickness of the structural member that is equal to a difference between the baseline thickness and the recess depth; and a reinforcing element intermediate the first surface and the

second surface; wherein the reinforcing element is disposed intermediate the recessed region and the first surface, supports a portion of the structural member having the effective thickness, and is selectively positioned relative to the first surface such that a resistance to a given bending moment for the structural member as a whole is optimized.

2. The structural member of claim 1 wherein a material used for the structural member and defining the first surface and the second surface is concrete.

3. The structural member of claim 2 wherein the reinforcing element comprises a steel bar.

4. The structural member of claim 3 wherein the reinforcing element comprises a lattice of interconnected steel bars.

5. The structural member of claim 2 wherein the reinforcing element comprises a web structure.

6. The structural member of claim 5 wherein the web structure comprises a steel mesh.

7. The structural member of claim 5 wherein the web structure comprises a fibrous material.

8. The structural member of claim 2 wherein the recessed region is symmetrical in planar cross-section.

9. The structural member of claim 8 wherein the recessed region is rectangular in planar cross-section.

10. The structural member of claim 8 wherein the recessed region is trapezoidal in planar cross-section.

11. The structural member of claim 8 wherein the recessed region is curved in planar cross-section.

12. A method employing a modular framework system to facilitate fabrication of a structural member, the method comprising: securing an elongate base plate to a foundation operative to support the structural member, the securing comprising selectively attaching the elongate base plate to the foundation in an orientation substantially perpendicular to a longitudinal axis of a footprint of the structural member; attaching a column to a portion of the elongate base plate proximate to the footprint of the structural member and bracing the column in a substantially vertical orientation using a brace member attached to a portion of the elongate base plate distal to the footprint of the structural member, the column having a height selected to accommodate a selected height of the structural member; attaching a top support to the column opposite the elongate base plate and securing the top support in an orientation substantially parallel to a direction of the elongate base plate; selectively attaching a rail assembly to the top support in a substantially horizontal orientation and substantially parallel to the longitudinal axis of the footprint of the structural member; selectively suspending an interior form element from the rail assembly to define an interior surface of the structural member; selectively suspending an exterior form element from the rail assembly a distance from the interior form element to define an exterior surface of the structural member, the distance between the interior form element and the exterior form element defining a baseline thickness of the structural member; optionally selectively suspending an opening form element to define an opening, in which no structural material will be present, in the structural member, the optionally selectively suspending comprising selectively attaching the opening form element to at least one of the interior form element or the exterior form element; selectively suspending a reinforcing element from the rail assembly intermediate the interior form element and the exterior form element and positioning the reinforcing element in a position relative to the exterior form element to support the structural member against a bending moment; and depositing a slurry of construction material in a cavity between the interior form element and the exterior form element to form the structural member having the selected height.

13. The method of claim 12 wherein the selectively suspending an interior form element comprises suspending a recess element, having a recess depth from the interior surface of the structural member, that creates a recessed region having an effective thickness that is less than the baseline thickness of the structural member when the slurry is deposited in the cavity.

14. The method of claim 13 wherein the recess element is rectangular in planar cross-section.

15. The method of claim 13 wherein the recess element is trapezoidal in planar cross-section.

- 16.** The method of claim 13 wherein the recess element is curved in planar cross-section.
- 17.** The method of claim 13 wherein the selectively suspending a reinforcing element comprises positioning the reinforcing element proximate to the external form element such that the reinforcing element is entirely encased within an area of the structural element having the effective thickness.
- 18.** The method of claim 12 wherein the depositing a slurry comprises depositing concrete in the cavity.
- 19.** The method of claim 18 further comprising utilizing a trolley that translates on the rail assembly along the longitudinal axis of the footprint of the structural member to deposit the concrete in the cavity.
- 20.** A modular framework system to facilitate fabrication of a structural member, the system comprising: an elongate base plate selectively attached to a foundation, which is operative to support the structural member, in an orientation substantially perpendicular to a longitudinal axis of a footprint of the structural member; a column attached to a portion of the elongate base plate proximate to the footprint of the structural member and having a height selected to accommodate a selected height of the structural member; a brace member attached to the column at a location above the foundation and below the height of the structural member and to a portion of the elongate base plate distal to the footprint of the structural member such that the brace member braces the column in a substantially vertical orientation during use; a top support attached to the column opposite the elongate base plate in an orientation substantially parallel to a direction of the elongate base plate; a rail assembly attached to the top support in a substantially horizontal orientation and substantially parallel to the longitudinal axis of the structural member; an interior form element selectively suspended from the rail assembly and defining an interior surface of the structural member; an exterior form element selectively suspended from the rail assembly a distance from the interior form element to define an exterior surface of the structural member, the distance between the interior form element and the exterior form element creating a cavity that defines a baseline thickness of the structural member; an opening form element optionally selectively suspended from at least one of the interior form element or the exterior form element to define an opening, in which no structural material will be present, in the structural member; and a reinforcing element selectively suspended from the rail assembly intermediate the interior form element and the exterior form element and positioned relative to the exterior form element to support the structural member against a bending moment.
- 21.** The system of claim 20 wherein the interior form element comprises a recess element, having a recess depth from the interior surface of the structural member, that creates a recessed region having an effective thickness that is less than the baseline thickness of the structural member when a slurry is deposited in the cavity.
- 22.** The system of claim 21 wherein the recessed region is symmetrical in planar cross-section.
- 23.** The system of claim 22 wherein the recessed region is rectangular in planar cross-section.
- 24.** The system of claim 22 wherein the recessed region is trapezoidal in planar cross-section.
- 25.** The system of claim 22 wherein the recessed region is curved in planar cross-section.
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